



**PHYSICS
WALLAH**

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Chemistry

CHAPTER

SOME BASIC CONCEPTS OF CHEMISTRY



Some Basic Concepts of Chemistry

BASICS OF CHEMISTRY

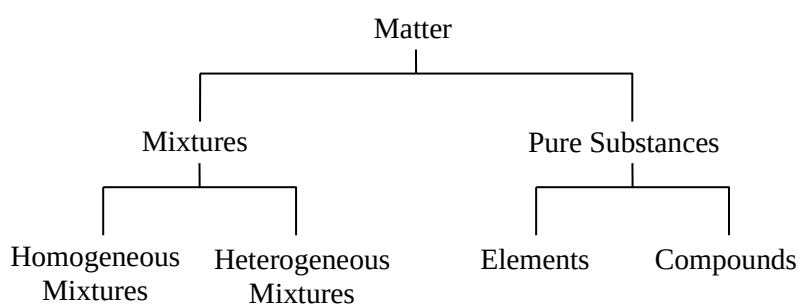
1. Classification of Matter:

Physical classification: It is based on physical state under ordinary conditions of temperature and pressure, matter is classified into the following three types:

- (a) Solid (b) Liquid (c) Gas

- (a) **Solid:** A substance is said to be solid if it possesses a definite volume and a definite shape
E.g. Sugar, iron, gold, wood etc.
- (b) **Liquid:** A substance is said to be liquid if it possesses a definite volume but not definite shape. They take up the shape of the vessel in which they are put.
E.g. Water, milk, oil, mercury, alcohol etc.
- (c) **Gas:** A substance is said to be gas if it neither possesses a definite volume nor a definite shape. This is because they fill up the whole vessel in which they are put.
E.g. Hydrogen(H_2), oxygen(O_2), carbon dioxide(CO_2), etc.

Chemical classification:



On the basis of chemical nature matter is classified into the following two types:

- (a) **Pure Substance**
(b) **Mixture**

- (a) **Pure Substance:** When all constituent particles of a substance are same in chemical nature, it is said to be a pure substance. Two type of pure substances:
- (i) **Element** (ii) **Compound**
- (i) **Element:** an element consists of only one type of atoms. These particles may exist as Atoms or molecules. E.g. O_2 , P_4 , S_8 , etc.

(ii) **Compound:** When two or more atoms of different elements combine together in a definite ratio. E.g. water, ammonia, carbon dioxide, sugar, etc.

(b) **Mixture:** A mixture contains many types of particles. A mixture contains particles of two or more pure substances which may be present in it in any ratio. Hence, their composition is variable. Pure substances forming mixture are called its components. Many of the substances present around you are mixtures.

For example, sugar solution in water, air, tea, etc., are all mixtures. Two types of mixtures:

(i) **Homogeneous mixture**

(ii) **Heterogeneous mixture**

(i) **Homogeneous mixture:** The components completely mix with each other. This means particles of components of the mixture are uniformly distributed throughout the bulk of the mixture and its composition is uniform throughout. E.g.: Sugar solution, air

(ii) **Heterogeneous mixture:** In a heterogeneous mixture, the composition is not uniform throughout and sometimes different components are visible. E.g.: mixtures of salt and sugar, grains and pulses along with some dirt (often stone pieces)

2. Introduction:

There are a large number of objects around us which we can see and feel. It was John Dalton who firstly developed a theory on the structure of matter, later on which is known as Dalton's atomic theory.

Dalton's Atomic Theory:

- (i) Each element is composed of extremely small particles called atoms which can take part in chemical combination.
- (ii) All atoms of a given element are identical i.e., atoms of a particular element are all alike but differ from atoms of other elements.
- (iii) Atoms of different elements possess different properties (including different masses).
- (iv) Atoms are indestructible i.e., atoms are neither created nor destroyed in chemical reactions.
- (v) Atoms of elements combine to form molecules and compounds are formed when atoms of more than one element combine.
- (vi) In a given compound, the relative number and kind of atoms is constant.

Atomic mass: It is the average relative mass of atom of element as compared with times the mass of an atom of carbon-12 isotope.

$$\text{Atomic mass} = \frac{\text{Average mass of an atom}}{1/12 \times \text{Mass of an atom of C}^{12}}$$

Gram atomic mass (GAM): Atomic mass of an element expressed in grams is called Gram atomic mass or gram atom or mole atom.

$$(i) \text{ Number of gram atoms} = \frac{\text{Mass of an element}}{\text{GAM}}$$

$$(ii) \text{ Mass of an element in g} = \text{No. of gram atoms} \times \text{GAM}$$

$$(iii) \text{ Number of atoms in 1 GAM} = 6.02 \times 10^{23}$$

$$\text{Number of atoms in a given substance} = \text{No. of gram atoms} \times 6.02 \times 10^{23}$$

$$(iv) \text{ Number of atoms in 1 g of element} = \frac{\text{Mass}}{\text{GAM}}$$

$$(v) \text{ Mass of one atom of the element (in g)} = \frac{\text{GAM}}{6.02 \times 10^{23}}$$

Molecular mass: Molecular mass of a molecule, of an element or a compound may be defined as a number which indicates how many times heavier is a molecule of that element or compound as compared with $\frac{1}{12}$ of the mass of an atom of carbon-12. Molecular mass is also expressed in a.m.u.

$$\text{Molecular mass} = \frac{\text{Mass of one molecule of the substance}}{1/12 \times \text{Mass of one atom of C-12}}$$

$$\text{Actual mass of one molecule} = \text{Mol. mass (in amu)} \times 1.66 \times 10^{-24} \text{ g}$$

Molecular mass of a substance is the additive property and can be calculated by adding the atomic masses of atoms present in one molecule.

Gram molecular mass (GMM): Molecular mass of an element or compound when expressed in g is called its gram molecular mass, gram molecule or mole molecule.

$$\text{Number of gram molecules} = \frac{\text{Mass of substance}}{\text{GMM}}$$

$$\text{Mass of substance in g} = \text{No. of gram molecules} \times \text{GMM}$$

Element	R.A.M. (Relative Atomic Mass)	Atomic mass (mass of one atom)	Gram Atomic mass /weight
N	14	14 amu	14 gm
He	4	4 amu	4 gm
C	12	12 amu	12 gm

Mole: One mole of any substance contains a fixed number (6.023×10^{23}) of any type of particles (atoms or molecules or ions) and has a mass equal to the atomic or molecular weight, in grams. Thus, it is correct to refer to a mole of helium, a mole of electrons or a mole of any ion, meaning respectively Avogadro's number of atoms, electrons or ions.

Methods of Calculations of mole:

(1) If no. of some species is given, then no. of moles = $\frac{\text{Given no}}{N_A}$

(2) If weight of a given species is given, then no. of moles = $\frac{\text{Given wt.}}{\text{Atomic wt.}}$ (for atoms),

or = $\frac{\text{Given wt.}}{\text{Molecular wt.}}$ (for molecules)

(3) If volume of a gas is given along with its temperature (T) and pressure (P).
 $PV = nRT$ (n: Number of moles of gas)

P (Pressure of gas): Pressure of the gas is the force exerted by the gas per unit area of the walls of the container in all directions.

$$\text{Thus, Pressure (P)} = \frac{\text{Force(F)}}{\text{Area(A)}} = \frac{\text{Mass(m)} \times \text{Acceleration(a)}}{\text{Area(A)}}$$

Name	Symbol	Value
Bar	bar	1 bar = 10^5 Pa
Atmosphere	atm	1 atm = 1.01325×10^5 Pa
Torr	Torr	1 Torr = $\frac{101325}{760}$ Pa $\times$ 133.322 Pa
millimeter of mercury	mm Hg	1 mm Hg = 133.322 Pa

V (Volume of gas): Volume is expressed in liters (L), milliliters (mL) or cubic centimeters (cm^3), cubic meters (m^3).

$$1 \text{ m}^3 = 10^3 \text{ dm}^3 = 10^6 \text{ cm}^3 = 10^6 \text{ mL} = 10^3 \text{ L}$$

T (Temperature of gas): S.I. unit of temperature is Kelvin (K).

$$T(\text{K}) = t(^{\circ}\text{C}) + 273.15$$

R (Universal gas constant): Values of $R = 0.082 \text{ LatmK}^{-1}\text{mol}^{-1}$

$$= 8.314 \text{ JK}^{-1}\text{mol}^{-1}$$

$$= 1.987 \text{ CalK}^{-1}\text{mol}^{-1}$$

Note: 1 mole of atom is also termed as 1 g atom

1 mole of ions is also termed as 1 g ion

1 mole of molecule is also termed as 1 gmolecule

Ex.1 How much time (in years) would it take to distribute one Avogadro number of wheat grains if 10^{10} grains are distributed each second?

Ans. 1.9×10^6 years (approx)

Sol. 10^{10} grains are distributed in 1 second

$$6.02 \times 10^{23} \text{ grains are distributed in } \frac{6.02 \times 10^{23}}{10^{10}} \text{ sec} = \frac{6.02 \times 10^{23}}{10^{10} \times 60 \times 60 \times 24 \times 365} \text{ years}$$

$$= 1.9 \times 10^6 \text{ years (approx.)}$$

Ex.2 How many atoms are there in 100 amu of He?

Ans. 25

Sol. We know that, $1 \text{ amu} = \frac{1}{12} \times \text{weight of one } ^{12}\text{C atom}$

or weight of one $^{12}\text{C atom} = 12 \text{ amu}$ (at. wt. of C = 12 amu).

Similarly, as the atomic weight of He is 4

weight of one He atom = 4 amu.

$$\text{Thus, the number of atoms in 100 amu of He} = \frac{100}{4} = 25.$$

Ex.3 The density of liquid mercury is 13.6 g/cm^3 . How many moles of mercury are there in 1 litre of the metal? (Atomic mass of Hg = 200.)

Ans. 68

Sol. 1 liter Hg metal

volume = 1000

$$d = \frac{m}{v} \quad \text{mass} = d \times V = 13.6 \times 1000$$

$$\text{No of mole of Hg metal} = \frac{13.6 \times 1000}{200} = 68 \text{ mole}$$

Ex.4 Calculate the number of molecules in a drop of water weighing 0.09 g.

Ans. 3.01×10^{21} molecules of H_2O

Sol. Number of mole = $\frac{0.09}{18}$

$$\text{so, number of molecules} = \frac{0.09}{18} \times N_A = 3.01 \times 10^{21}.$$

Ex.5 A sample of ethane has the same mass as 10.0 million molecules of methane. How many C_2H_6 molecules does the sample contain?

Ans. 5.34×10^6

Sol. Let the number of C_2H_6 molecules in the sample be n . As given, mass of C_2H_6 = mass of 10^7 molecules of CH_4

$$\frac{n}{N_A} \times \text{mol. wt. of } C_2H_6 = \frac{10^7}{N_A} \times \text{mol. wt. of } CH_4$$

$$\frac{n}{N_A} \times 30 = \frac{10^7}{N_A} \times 16$$

$$n = 5.34 \times 10^6.$$

Ex.6 If, from 10 moles NH_3 and 5 moles of H_2SO_4 , all the H-atoms are removed in order to form H_2 gas, then find the number of H_2 molecules formed.

Ans. $20 N_A$

Sol. 10 mole NH_3 have mole of 'H' atom = 10×3

5 mole of H_2SO_4 have mole of 'H' atom = 10

Total mole of 'H' atom = 40

mole of H_2 = 20

Ex.7 The weight of 350 mL of a diatomic gas at $0^\circ C$ and 2 atm pressure is 1 g. The weight of one atom is:

Ans. 16 amu

Sol. $PV = nRT$; $n = \frac{PV}{RT}$

$$n = \frac{2 \times 0.350}{0.0821 \times 273}$$

$$n = \frac{\text{Weight}}{\text{Atomic mass}}$$

$$\text{Atomic mass} = 16 \text{ amu}$$

Ex.8 Oxygen is present in a 1-litre flask at a pressure of 7.6×10^{-10} mm of Hg at $0^\circ C$. Calculate the number of oxygen molecules in the flask.

Ans. 0.44×10^{-13}

Sol. Pressure = 7.6×10^{-10} mm $\Rightarrow 0.76 \times 10^{-10}$ cm = atm (1 atm = 76 cm) = 10^{-12} atm.

Volume = 1 litre, $R = 0.0821$ lit. atm/K/mole, temperature = 273 K.

We know that $PV = nRT$ or $n = \frac{pV}{RT}$

$$n = \frac{10^{-12} \times 1}{0.082 \times 273} = 0.44 \times 10^{-13}$$

Ex.9 Equal volumes of oxygen gas and a second gas weigh 1.00 and 2.375 grams respectively under the same experimental conditions. Which of the following is the unknown gas?

(A) NO (B) SO_2 (C) CS_2 (D) CO

Ans. (C)

Sol. Moles of O_2 = Moles of X(unknown gas)

$$\frac{1}{32} = \frac{2.375}{M_x} \Rightarrow M_x = 76$$

Do yourself-1

- The number of molecules in 16 g of methane is:
- A sample of aluminium has a mass of 54.0 g. What is the mass of the same number of magnesium atoms? (At. wt. Al = 27, Mg = 24)
(A) 12 g (B) 24 g (C) 48 g (D) 96 g
- Find the total number of H, S and 'O' atoms in the following:
(i) 196 gm H_2SO_4 (ii) 196 amu H_2SO_4
(iii) 5 mole $\text{H}_2\text{S}_2\text{O}_8$ (iv) 3 molecules $\text{H}_2\text{S}_2\text{O}_6$
- The volume of a gas at 0°C and 700 mm pressure is 760 cc. The number of molecules present in this volume is:
- Four 1-litre flasks are separately filled with the gases H_2 , He, O_2 and O_3 at the same temperature and pressure. The ratio of total number of atoms of these gases present in different flask would be:
(A) 1 : 1 : 1 : 1 (B) 1 : 2 : 2 : 3 (C) 2 : 1 : 2 : 3 (D) 3 : 2 : 2 : 1
- The weight of 2.01×10^{23} molecules of CO is:
(A) 9.3 g (B) 7.2 g (C) 1.2 g (D) 3 g
- How many moles of e^- weight one Kg:
(A) 6.023×10^{23} (B) $\frac{1}{9.108} \times 10^{31}$ (C) $\frac{6.023}{9.108} \times 10^{54}$ (D) $\frac{1}{9.108 \times 6.023} \times 10^8$

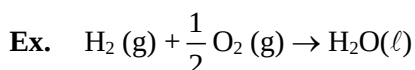
3. Laws of Chemical Combination:

Antoine Lavoisier, John Dalton and other scientists formulate certain laws concerning the composition of matter and chemical reactions. These laws are known as the law of chemical combination.

The Law of Conservation of Mass:

It is given by Antoine Lavoisier.

In a chemical change total mass remains conserved that is mass before reaction is always equal to mass after reaction.



Before reaction initially 1 mole $\frac{1}{2}$ mole

After the reaction 0 0 1 mole

mass before reaction = mass of 1 mole $\text{H}_2(\text{g})$ + mass of $\frac{1}{2}$ mole $\text{O}_2(\text{g})$

= 2 + 16 = 18 g

mass after reaction = mass of 1 mole water = 18 g

Law of Constant or Definite Proportion:

It is given by Proust. All chemical compounds are found to have constant composition irrespective of their method of preparation or sources.

Ex. In water (H_2O), Hydrogen and Oxygen combine in 2 : 1 molar ratio, the ratio remains constant whether it is tap water, river water or sea water or produced by any chemical reaction.

Ex.10 1.80 g of a certain metal burnt in oxygen gave 3.0 g of its oxide. 1.50 g of the same metal heated instead gave 2.50 g of its oxide. Show that these results illustrate the law of constant proportion.

Sol. In the first sample of the oxide,

wt. of metal = 1.80 g,

wt. of oxygen = (3.0 – 1.80) g

= 1.2 g

$$\frac{\text{wt. of metal}}{\text{wt. of oxygen}} = \frac{1.80\text{g}}{1.2\text{g}} = 1.5$$

In the second sample of the oxide,

$$\frac{\text{wt. of metal}}{\text{wt. of oxygen}} = \frac{1.50\text{g}}{1\text{g}} = 1.5$$

Thus, in both samples of the oxide the proportions of the weights of the metal and oxygen are fixed. Hence, the results follow the law of constant proportion.

The Law of Multiple Proportions:

It is given by Dalton.

When one element combines with the other element to form two or more different compounds, the mass of one element, which combines with a constant mass of the other, bear a simple ratio to one another.

Note: Simple ratio here means the ratio between small natural numbers, such as 1: 1, 1: 2, 1: 3, Later on this simple ratio becomes the valency and then oxidation state of the element.

Ex. Carbon and Oxygen when combine, can form two oxides, CO (carbon monoxide), CO₂(Carbon dioxides)

In CO, 12 g carbon combined with 16 g of oxygen.

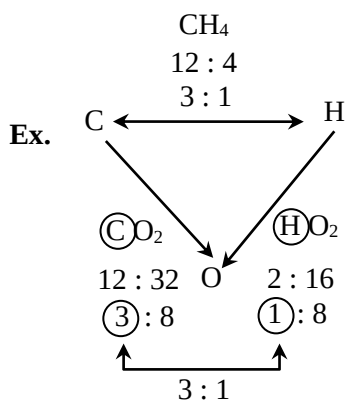
In CO₂, 12 g carbon combined with 32 g of oxygen.

Thus, we can see the mass of oxygen which combine with a constant mass of carbon (12 g) bear simple ratio of 16 : 32 or 1 : 2.

Law of Reciprocal Proportion:

It is given by Richter.

The ratio of the weights of two elements A and B which combine separately with a fixed weight of the third element C is either the same or simple ratio of the weights in which A and B combine directly with each other.



Ex.11 Ammonia contains 82.35% of nitrogen and 17.65% of hydrogen. Water contains 88.90% of oxygen and 11.10% of hydrogen. Nitrogen trioxide contains 63.15% of oxygen and 36.85% of nitrogen. Show that these data illustrate the law of reciprocal proportions.

Sol. In NH_3 , 17.65g of H combine with N = 82.35g

$$1 \text{ g of H combine with N} = \frac{82.35}{17.65} \text{ g} = 4.67 \text{ g}$$

In H_2O , 11.10 g of H combine with O = 88.90 g

$$1 \text{ g of H combine with O} = \frac{88.90}{11.10} \text{ g} = 8.01 \text{ g}$$

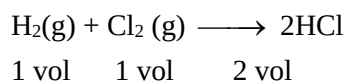
Ratio of the weights of N and O which combine with fixed weight (=1g) of H
 $= 4.67 : 8.01 = 1 : 1.7$

In N_2O_3 , ratio of weights of N and O which combine with each other = 36.85 : 63.15
 $= 1 : 1.7$

Thus, the two ratios are the same. Hence it illustrates the law of reciprocal proportions.

Gay-Lussac's Law of Combining Volume:

According to him elements combine in a simple ratio of atoms, gases combine in a simple ratio of their volumes provided all measurements should be done in the same temperature and pressure



Avogadro's Hypothesis:

Equal volumes of polyatomic all gases have equal number of molecules (not atoms) at same temperature and pressures conditions.

S.T.P. (Standard Temperature and Pressure)

At S.T.P. condition:

Temperature = 0°C or 273 K

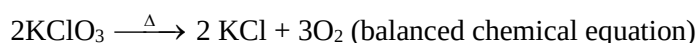
Pressure = 1 bar

and volume of one mole of gas at STP is found to be equal to 22.7litres which is known as molar volume.

4. Stoichiometry and Stoichiometric Calculations:

The word 'stoichiometry' is derived from two Greek words — *stoicheion* (meaning, *element*) and *metron* (meaning, *measure*). Stoichiometry, thus, deals with the calculation of masses (sometimes volumes also) of the reactants and the products involved in a chemical reaction

Ex. When potassium chlorate (KClO_3) is heated it gives potassium chloride (KCl) and oxygen (O_2).



$\Rightarrow$ Remember a balanced chemical equation is one which contains an equal number atoms of each element on both sides of equation.

Interpretation of balanced chemical equations:

Once we get a balanced chemical equation then we can interpret a chemical equation by following ways

(a) Mass - mass analysis

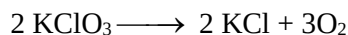
(b) Mass - volume analysis

(c) Mole - mole analysis

(a) Mole - mole analysis:

This analysis is very much important for quantitative analysis point of view.

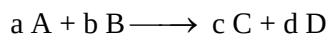
Now consider again the decomposition of KClO_3 .



In very first step of mole-mole analysis you should read the balanced chemical equation like **2 moles KClO_3 on decomposition gives you 2 moles KCl and 3 moles O_2** and from the stoichiometry of reaction we can write

$$\frac{\text{Moles of } \text{KClO}_3}{2} = \frac{\text{Moles of } \text{KCl}}{2} = \frac{\text{Moles of } \text{O}_2}{3}$$

Now for any general balance chemical equation like



you can write.

$$\frac{\text{Moles of A reacted}}{a} = \frac{\text{Moles of B reacted}}{b} = \frac{\text{Moles of C produced}}{c} = \frac{\text{Moles of D produced}}{d}$$

(b) Mass - mass analysis:

Consider the reaction $2 \text{KClO}_3 \longrightarrow 2 \text{KCl} + 3 \text{O}_2$ According to stoichiometry of the reaction mass-mass ratio: $2 \times 122.5 : 2 \times 74.5 : 3 \times 32$

$$\text{or } \frac{\text{Mass of } \text{KClO}_3}{\text{Mass of } \text{KCl}} = \frac{2 \times 122.5}{2 \times 74.5} \qquad \frac{\text{Mass of } \text{KClO}_3}{\text{Mass of } \text{O}_2} = \frac{2 \times 122.5}{3 \times 32}$$

Ex.12 Consider the balanced reaction,



What can be concluded from the coefficients of species in this balanced equation?

- (A) For this reaction, exactly 2 g of Cl_2O_7 must be taken to start the reaction
- (B) For this reaction, exactly 2 mol of Cl_2O_7 must be taken to start the reaction
- (C) Mole ratio of Cl_2O_7 , ClO_2 and O_2 during a chemical reaction at any instant (excluding any negative sign) are 2, 4 and 3 respectively
- (D) The ratio of change in number of moles of Cl_2O_7 , ClO_2 and O_2 is 2 : 4 : 3 (excluding any negative sign)

Ans. (D)

Sol. It follows directly from definition of stoichiometry.

Ex.13 Calculate the weight of iron which will be converted into its oxide by the action of 36 g of steam.



Ans. 84 g

Sol. Mole ratio of reaction suggests,

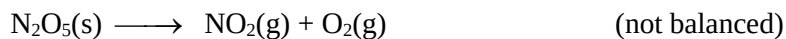
$$\text{Mole of Fe} = \frac{3}{4} \qquad \text{mol of H}_2\text{O}$$

$$= \frac{3}{4} \times \frac{36}{18} = \frac{3}{2}$$

$$\text{wt. of Fe} = \frac{3}{2} \times 56 = 84 \text{ g}$$

Ex.14 When Dinitrogen pentaoxide (N_2O_5 , a white solid) is heated, it decomposes into nitrogen dioxide and oxygen.

If a sample of N_2O_5 produces 1.6 g O_2 , then how many grams of NO_2 are formed?



- (A) 9.2 g (B) 4.6 g (C) 2.3 g (D) 18.4 g

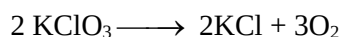
Ans. (A)

Sol. $\text{N}_2\text{O}_5(\text{s}) \longrightarrow 2\text{NO}_2(\text{s}) + \frac{1}{2}\text{O}_2$ (Balanced reaction)

$$\frac{\text{Mole of O}_2}{1/2} = \frac{\text{Mole of NO}_2}{2} \Rightarrow \frac{1.6}{32} \times 2 \times 2 = \text{Mole of NO}_2 = 0.2 \Rightarrow \text{wt of NO}_2 = 0.2 \times 46 = 9.2 \text{ g}$$

(c) Mass - volume analysis:

Now again consider decomposition of KClO_3



mass volume ratio: $2 \times 122.5 \text{ g} : 2 \times 74.5 \text{ g} : 3 \times 22.4 \text{ L}$ at STP

we can use two relation for volume of oxygen

$$\frac{\text{Mass of KClO}_3}{\text{volume of O}_2 \text{ at STP}} = \frac{2 \times 122.5 \text{ g}}{3 \times 22.4 \text{ L}} \quad \dots(\text{i})$$

$$\text{and } \frac{\text{Mass of KCl}}{\text{volume of O}_2 \text{ at STP}} = \frac{2 \times 74.5 \text{ g}}{3 \times 22.4 \text{ L}} \quad \dots(\text{ii})$$

Ex.15 When oxygen gas is passed through Siemen's ozoniser, it completely gets converted into ozone gas. The volume of ozone gas produced at 1 atm and 273K, if initially 96 g of oxygen gas was taken, is:

- (A) 44.8 L (B) 89.6 L (C) 67.2 L (D) 22.4 L

Ans. (A)

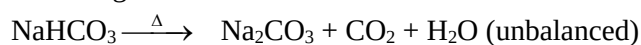
Sol. $3\text{O}_2 \longrightarrow 2\text{O}_3$

$$\text{Mole} = \frac{96}{32} = 3 \quad \text{mole} = 2$$

$$\text{Volume of O}_3 \text{ gas at 1 atm and 273K} = 2 \times 22.4 = 44.8 \text{ L}$$

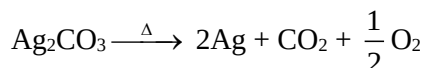
Do yourself-2

1. Assuming 100% yield of the reaction, how many moles of NaHCO_3 will produce 448 mL of CO_2 gas at STP according to the reaction:



- (A) 0.04 (B) 0.4 (C) 4 (D) 40

2. Calculate the residue obtained on strongly heating 2.76 g Ag_2CO_3 .



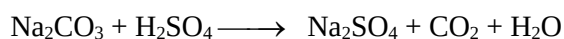
3. For the reaction $2\text{P} + \text{Q} \longrightarrow \text{R}$, 4 mol of P and excess of Q will produce:

- (A) 8 mol of R (B) 5 mol of R (C) 2 mol of R (D) 1 mol of R

4. If 1.5 moles of oxygen combine with Al to form Al_2O_3 , the weight of Al used in the reaction is:

- (A) 27 g (B) 40.5 g (C) 54g (D) 81 g

5. How many liters of CO_2 at STP will be formed when 0.01 mol of H_2SO_4 reacts with excess of Na_2CO_3 .



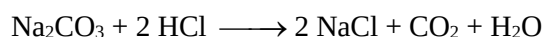
- (A) 22.7 L (B) 2.27 L (C) 0.227 L (D) 1.135 L

Limiting Reagent:

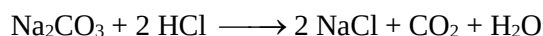
Many a time, reactions are carried out with the amounts of reactants that are different than the amounts as required by a balanced chemical reaction. In such situations, one reactant is in more amount than the amount required by balanced chemical reaction. The reactant which is present in the least amount gets consumed after sometime and after that further reaction does not take place whatever be the amount of the other reactant. Hence, the reactant, which gets consumed first, limits the amount of product formed and is, therefore, called the limiting reagent.

The reactant which consumed first into the reaction when we are dealing with balance chemical equation then if number of moles of reactants are not in the ratio of stoichiometric coefficient of balanced chemical equation, then there should be one reactant which should be limiting reactant.

Ex.16 Six mole of Na_2CO_3 is reacted with 4 moles of HCl solution. Find the volume of CO_2 gas produced at STP. The reaction is:



Sol. From the reaction:



gives moles

3 mol

6 mol

given mole ratio

1

: 2

Stoichiometric coefficient ratio

1

: 2

See here given number of moles of reactants are not in stoichiometric coefficient ratio. Therefore there should be one reactant which consumed first and becomes limiting reagent.

But the question is how to find which reactant is limiting, it is not very difficult you can easily find it. According to the following method.

How to find limiting reagent:**Step: I**

Divide the given moles of reactant by the respective stoichiometric coefficient of that reactant.

Step: II

See for which reactant this division comes out to be minimum. The reactant having minimum value is limiting reagent.

Step: III

Now once you find limiting reagent then your focus should be on limiting reagent

From

Step I & II

Na_2CO_3

HCl

$$\frac{6}{1} = 6$$

$$\frac{4}{2} = 2 \text{ (Division in minimum)}$$

$\therefore$ HCl is limiting reagent

From Step III

$$\text{From } \frac{\text{Mole of HCl}}{2} = \frac{\text{Mole of CO}_2 \text{ produced}}{1}$$

$\therefore$ Mole of CO_2 produced = 2 moles

$\therefore$ Volume of CO_2 produced at S.T.P. = $2 \times 22.7 = 45.4 \text{ L}$

Ex.17 In the reaction $4\text{A} + 2\text{B} + 3\text{C} \longrightarrow \text{A}_4\text{B}_2\text{C}_3$ what will be the number of moles of product formed, starting from 2 moles of A, 1.2 moles of B & 1.44 moles of C:

(A) 0.5

(B) 0.6

(C) 0.48

(D) 4.64

Ans. (C)

Sol.



Initial mole

2

1.2

1.44

0

Final mole

0

0.48

C is limiting reagent.

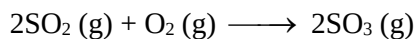
Moles of $\text{A}_4\text{B}_2\text{C}_3$ is 0.48.

Ex.18 A 5 g mixture of SO_2 and O_2 gases is reacted to form SO_3 gas. What should be the mass ratio of SO_2 and O_2 gases in mixture to obtain maximum amount of SO_3 gas:

- (A) 4 : 1 (B) 3 : 2
(C) 2 : 3 (D) 1 : 4

Ans. (A)

Sol. For maximum amount of product, the reactants should be present in their stoichiometric ratio.



mass x 5 - x

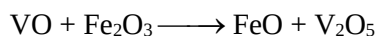
mole $\frac{x}{64}$ $\frac{5-x}{32}$

$$\text{So, } \frac{\left(\frac{x}{64}\right)}{\left(\frac{5-x}{32}\right)} = 2 : 1$$

Therefore, x = 4

$$m_{\text{SO}_2} : m_{\text{O}_2} = 4 : 1$$

Ex.19 Calculate the weight of FeO from 4 g VO and 5.75 g of Fe_2O_3 . Also report the limiting reactant.



Ans. Weight of FeO formed = 5.17 g

Sol. Balanced equation 2VO + $3\text{Fe}_2\text{O}_3$ $\longrightarrow$ 6FeO + V_2O_5

Moles before reaction $\frac{4}{67}$ $\frac{5.75}{160}$ 0 0

= 0.05970 0.03590

Moles after reaction $(0.05970 - 0.0359)$ 0 $\left(\frac{6}{5} \times 0.0359\right)$ $\left(\frac{1}{3} \times 0.0359\right)$

As 2 moles of VO react with 3 moles of Fe_2O_3

$$\therefore 0.05970 \text{ g moles of VO} = \frac{3}{2} 0.05970 = 0.08955 \text{ moles of } \text{Fe}_2\text{O}_3$$

Moles of Fe_2O_3 available = 0.0359 only

Hence, Fe_2O_3 is the limiting reagent.

$$\text{Moles of FeO formed} = \frac{6}{3} \times 0.0359$$

$$\therefore \text{Weight of FeO formed} = 0.0359 \times 2 \times 72 = 5.17 \text{ g}$$

$$\left(\frac{n_{\text{FeO}}}{n_{\text{Fe}_2\text{O}_3}} = \frac{6}{3}\right) \Rightarrow n_{\text{FeO}} = \frac{6}{3} \times n_{\text{Fe}_2\text{O}_3}$$

Do yourself-3

- The reaction
 $2C + O_2 \longrightarrow 2CO$
 is carried out by taking 24 g of carbon and 128 g of O_2 .
 Find out:
 (i) Which reactant is left in excess?
 (ii) How much of it is left?
 (iii) How many moles of CO are formed?
 (iv) How many grams of other reactant should be taken so that nothing is left at the end of reaction?
- How many mole of $Zn(FeS_2)$ can be made from 2 mole zinc, 3 mole iron and 5 mole sulphur.
 (A) 2 mole (B) 3 mole (C) 4 mole (D) 5 mole
- Calculate the amount of Ni needed in the Mond's process given below:
 $Ni + 4CO \longrightarrow Ni(CO)_4$
 If CO used in this process is obtained through a process, in which 6 g of carbon is mixed with 44 g CO_2 .
 (Ni = 59 u)
 (A) 14.675 g (B) 29 g (C) 58 g (D) 28 g
- 0.5 mole of H_2SO_4 is mixed with 0.2 mole of $Ca(OH)_2$. The maximum number of moles of $CaSO_4$ formed is:
 (A) 0.2 (B) 0.5 (C) 0.4 (D) 1.5
- The mass of Na_2SO_4 produced from 196 gram of H_2SO_4 and 1 mole of NaOH:
 (A) 49 g (B) 98 g (C) 61 g (D) 34.3 g

Principle of Atom Conservation (Poac):

Infect POAC is nothing but the conservation of mass, expressed before in the concepts of atomic theory and if atoms are conserved, moles of atoms shall also be conserved.

The principle is fruitful for the students when they don't get the idea of balanced chemical equation in the problem.

This principle can be under stand by the following example.

Consider the decomposition of $KClO_3(s) \rightarrow KCl(s) + O_2(g)$ (unbalanced chemical reacts ion)

Apply the principle of atom conservation (POAC) for K atoms.

Moles of K atoms in reactant = moles of K atoms in products

or moles of K atoms in $KClO_3$ = moles of K atoms in KCl

Now, since 1 molecule of $KClO_3$ contains 1 atom of K

or 1 mole of $KClO_3$ contains 1 mole of K, similarly 1 mole of KCl contains 1 mole of K

Thus, moles of K atoms in $KClO_3 = 1 \times$ moles of $KClO_3$

and moles of K atoms in KCl = $1 \times$ moles of KCl

$\therefore$ moles of $KClO_3$ = moles of KCl

or $\frac{\text{wt. of } KClO_3 \text{ in g}}{\text{mol. wt. of } KClO_3} = \frac{\text{wt. of KCl in g}}{\text{mol. wt. of KCl}}$

* The above equation gives the mass-mass relationship between $KClO_3$ and KCl which is important in stoichiometric calculations.

Again, applying the principle of atom conservation for O atoms,

moles of O in $\text{KClO}_3 = 3 \times \text{moles of KClO}_3$

moles of O in $\text{O}_2 = 2 \times \text{moles of O}_2$

$\therefore 3 \times \text{moles of KClO}_3 = 2 \times \text{moles of O}_2$

$$\text{or } \frac{\text{wt. of KClO}_3}{\text{mol. wt. of KClO}_3} = 2 \times \frac{\text{vol. of O}_2 \text{ at NTP}}{\text{standard molar vol. (22.4 l)}}$$

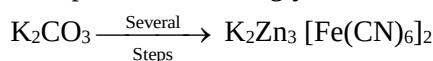
* The above equations thus give the mass-volume relationship of reactants and products.

Ex.20 27.6g K_2CO_3 was treated by a series of reagents so as to convert all of its carbon to $\text{K}_2\text{Zn}_3 [\text{Fe}(\text{CN})_6]_2$. Calculate the weight of the product.

[mol. wt. of $\text{K}_2\text{CO}_3 = 138$ and mol. wt. of $\text{K}_2\text{Zn}_3 [\text{Fe}(\text{CN})_6]_2 = 698$]

Ans. 11.6 g

Sol. Here we have not knowledge about series of chemical reactions but we known about initial reactant and final product accordingly



Since C atoms are conserved, applying POAC for C atoms,

Moles of C in $\text{K}_2\text{CO}_3 = \text{moles of C in K}_2\text{Zn}_3 [\text{Fe}(\text{CN})_6]_2$

$1 \times \text{moles of K}_2\text{CO}_3 = 12 \times \text{moles of K}_2\text{Zn}_3 [\text{Fe}(\text{CN})_6]_2$

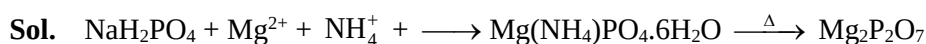
($\because$ 1 mole of K_2CO_3 contains 1 moles of C)

$$\frac{\text{wt. of K}_2\text{CO}_3}{\text{mol. wt. of K}_2\text{CO}_3} = 12 \frac{\text{wt. of the product}}{\text{mol. wt. of product}}$$

$$\text{wt. of K}_2\text{Zn}_3 [\text{Fe}(\text{CN})_6]_2 = \frac{27.6}{138} \times \frac{698}{12} = 11.6 \text{ g}$$

Ex.21 In a gravimetric determination of P of an aqueous solution of dihydrogen phosphate in H_2PO_4^- is treated with a mixture of ammonium and magnesium ions to precipitate magnesium ammonium phosphate, $\text{Mg}(\text{NH}_4)\text{PO}_4 \cdot 6\text{H}_2\text{O}$. This is heated and decomposed to magnesium pyrophosphate, $\text{Mg}_2\text{P}_2\text{O}_7$. A solution of H_2PO_4^- yielded 2.054 g of ($\text{Mg}_2\text{P}_2\text{O}_7$). What weight of NaH_2PO_4 was present originally?

Ans. 2.22 g



As P atoms are conserved, applying POAC for P atoms, moles of P in $\text{NaH}_2\text{PO}_4 = \text{Moles of P in Mg}_2\text{P}_2\text{O}_7$

$\Rightarrow 1 \times \text{Moles of NaH}_2\text{PO}_4 = 2 \times \text{Moles of Mg}_2\text{P}_2\text{O}_7$

$$\therefore \frac{W_{\text{NaH}_2\text{PO}_4}}{M_{\text{NaH}_2\text{PO}_4}} = 2 \times \frac{W_{\text{Mg}_2\text{P}_2\text{O}_7}}{M_{\text{Mg}_2\text{P}_2\text{O}_7}} \Rightarrow \frac{W_{\text{NaH}_2\text{PO}_4}}{120} = 2 \times \frac{2.054}{222}$$

$$\therefore W_{\text{NaH}_2\text{PO}_4} = 2.22 \text{ g}$$

Percentage yield:

In general, in any chemical reaction, the amount of product formed is always less than the calculated amount due to reversibility in the chemical reaction. Therefore, yield of a chemical reaction (Y) comes into picture and is given by:

$$\text{The percentage yield of product} = \frac{\text{actual yield}}{\text{the theoretical maximum yield}} \times 100$$

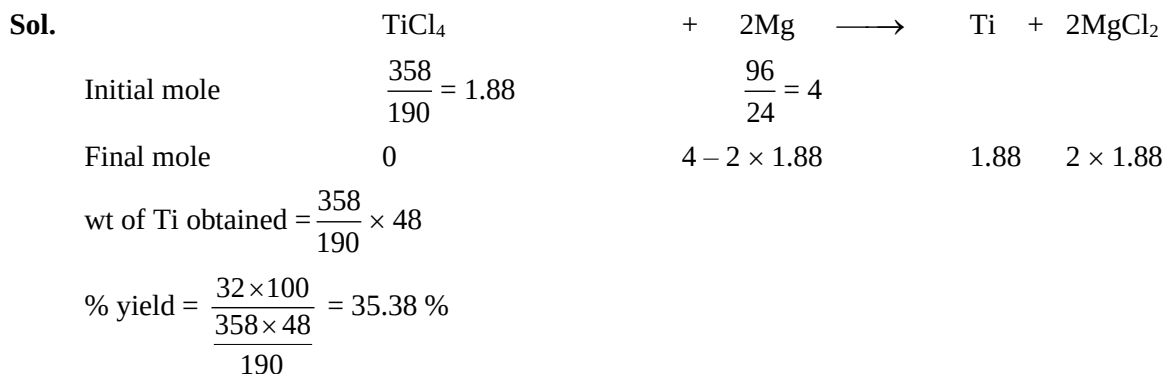
Ex.22 In a certain operation 358 g of TiCl_4 is reacted with 96 g of Mg. Calculate % yield of Ti if 32 g of Ti is actually obtained [At. wt. Ti = 48, Mg = 24]

(A) 35.38 %

(B) 66.6 %

(C) 100 %

(D) 60 %

Ans. (A)

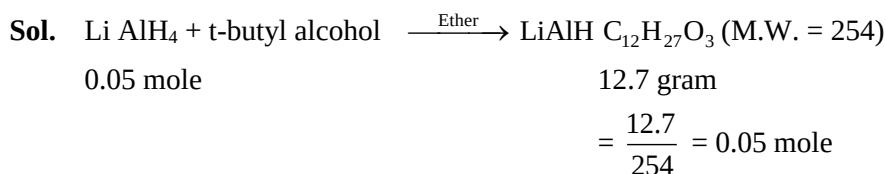
Ex.23 0.05 mole of LiAlH_4 in ether solution was placed in a flask containing 74g (1 mole) of t-butyl alcohol. The product $\text{LiAlHC}_{12}\text{H}_{27}\text{O}_3$ weighed 12.7 g. If Li atoms are conserved, the percentage yield is:
(Li = 7, Al = 27, H = 1, C = 12, O = 16).

(A) 25%

(B) 75%

(C) 100%

(D) 15%

Ans. (C)

Li atom remain conserved so

No. of mole of LiAlH_4 = No. of mole of $\text{LiAlHC}_{12}\text{H}_{27}\text{O}_3$ So, No. of mole of $\text{LiAlHC}_{12}\text{H}_{27}\text{O}_3 = 0.05$

$$\% \text{ yield} = \frac{0.05}{0.05} \times 100 = 100\%$$

Percentage Purity:

Ex.24 How much marble of 90.5% purity would be required to prepare 10 litres of CO_2 at 1 atm, 0°C when the marble is acted upon by dilute HCl?

Ans. 49.326 g

100 g 22.4 litre

22.4 L of CO_2 at STP will be obtained from 100 g of CaCO_3

$$\therefore 10 \text{ L of } \text{CO}_2 \text{ will be obtained from pure } \text{CaCO}_3 = \frac{100}{22.4} \times 10 = 44.64 \text{ g}$$

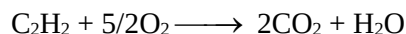
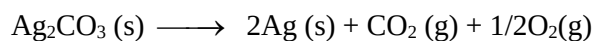
$$\therefore \text{Impure marble required} = \frac{100}{90.5} \times 44.64 = 49.326 \text{ g}$$

Sequential Reactions:

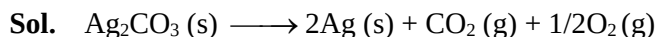
This reaction is defined as that reaction which proceeds from reactants to final products through one or more intermediate stages. The overall reaction is a result of several successive or consecutive steps.

Example: $\text{A} \rightarrow \text{B} \rightarrow \text{C}$

Ex.25 Minimum amount of Ag_2CO_3 (s) required to produce sufficient oxygen for the complete combustion of C_2H_2 which produces 11.2 L of CO_2 at 1 atm and 273K after combustion is: [Ag = 108]



Ans. 345 g



By Stoichiometry of reaction

$$\text{Moles of CO}_2 \text{ formed} = \frac{11.2}{22.4} = \frac{1}{2}$$

$$\text{Moles of O}_2 \text{ required} = \frac{5}{4} \times \frac{1}{2} = \frac{5}{8}$$

$$\text{Moles of Ag}_2\text{CO}_3 \text{ required} = 2 \times \frac{5}{8} = \frac{5}{4}$$

$$\text{Mass of Ag}_2\text{CO}_3 \text{ required} = \frac{5}{4} \times 276 = 345 \text{ g}$$

Ex.26 NX is produced by the following step of reactions,



How much M (metal) is consumed to produce 206 g of NX.

(Take at wt of M = 56, N=23, X = 80)

(A) 42 g (B) 56 g (C) 52g (D) 64 g

Ans. (A)



$$\text{Mole of NX} = \frac{206}{103} = 2$$

POAC for X Atom:

No. of X atom in M_3X_8 = No. of X Atom in NX

$$8 [\text{No. of mole of } \text{M}_3\text{X}_8] = 1 [\text{No. of mole of NX}]$$

$$\text{No. of mole of } \text{M}_3\text{X}_8 = \left[\frac{2}{8} \right] = \frac{1}{4} \text{ mole}$$

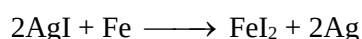
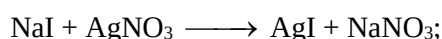
Now POAC for M Atom

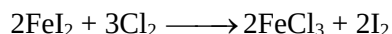
$$3 [\text{No. of mole of } \text{M}_3\text{X}_8] = 1 \times [\text{No. of Mole of M}]$$

$$\therefore 3 \times \frac{1}{4} = \text{No. of mole of M}$$

$$\text{weight of M atom} = \frac{3}{4} \times 56 = 42 \text{ gram}$$

Ex.27 The following process has been used to obtain iodine from oil-field brines in California.



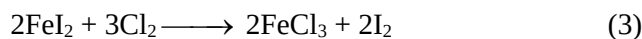
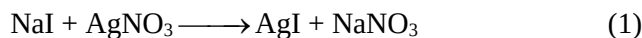


How many grams of AgNO_3 are required in the first step for every 254 kg I_2 produced in the third step:

- (A) 340 kg (B) 85 kg (C) 68 kg (D) 380 kg

Ans. (A)

Sol. Balanced equation:



From (3)

$$\frac{\text{mole of I}_2}{2} = \frac{\text{mole of FeI}_2}{2}$$

$$\frac{\text{mole of FeI}_2}{1} = \frac{\text{mole of AgI}}{2}$$

$$\frac{\text{mole of AgI}}{1} = \frac{\text{mole of AgNO}_3}{1}$$

$$\therefore \text{mole of I}_2 = (\text{mole of FeI}_2) \Rightarrow \left(\frac{\text{mole of AgI}}{2} \right) = \left(\frac{\text{mole of AgNO}_3}{2} \right)$$

$$\frac{254 \times 10^3}{254} = \frac{\text{mole of AgNO}_3}{2}$$

$$2 \times 10^3 = \text{mole of AgNO}_3 = \frac{\text{mass of AgNO}_3}{\text{molar mass of AgNO}_3}$$

$$\text{mass of AgNO}_3 = 170 \times (2 \times 10^3) \text{ g} = 340 \times 10^3 \text{ g} = 340 \text{ kg}.$$

Parallel Reactions:

The reactions in which a substance reacts or decomposes in more than one way are called parallel or side reactions.

Example: $\text{A} \longrightarrow \text{B}$

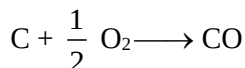
$\text{A} \longrightarrow \text{C}$

Ex.28 Find out moles of CO_2 & CO produced by combustion of 2 mol carbon with 1.25 O_2 leaving number residue:

Ans. $\text{CO}_2 = 0.5 \text{ mol}$, $\text{CO}_2 = 1.5 \text{ mol}$

Sol. $\text{C} + \text{O}_2 \longrightarrow \text{CO}_2$

x x x



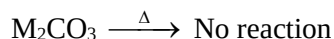
$$2-x \quad \frac{2-x}{2} \quad \quad \quad 2-x$$

$$x + 1 - \frac{x}{2} = 1.25$$

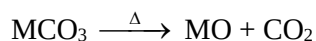
$$\frac{x}{2} = \frac{125}{100}$$

$$x = 0.5 \text{ mol}, \quad \text{CO}_2 = 0.5 \text{ mol}, \quad \text{CO}_2 = 1.5 \text{ mol}$$

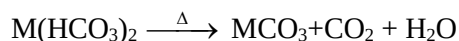
Mixture Analysis:

$$\text{Li}_2\text{CO}_3 \xrightarrow{\Delta} \text{Li}_2\text{O} + \text{CO}_2$$


(ii) All the carbonates of alkaline earth metals are thermally unstable and decompose on heating as follow.



(iii) Bicarbonates of both alkali metals and alkaline earth metals are decomposed at relatively low temperature as follow.



Ex.29 A sample of 3 g containing Na_2CO_3 and NaHCO_3 loses 0.248 g when heated to 300°C , the temperature at which NaHCO_3 decomposes to Na_2CO_3 , CO_2 and H_2O . What is the percentage of Na_2CO_3 in the given mixture?

Ans. 77.48%

Sol. The loss in weight is due to removal of CO_2 and H_2O which escape out on heating.

$$\text{wt. of Na}_2\text{CO}_3 \text{ in the product} = 3.00 - 0.248 = 2.752 \text{ g}$$

Let wt. of Na_2CO_3 in the mixture be x g

$$\therefore \text{wt. of NaHCO}_3 = (3.00 - x) \text{ g}$$

Since Na_2CO_3 in the products contains x g of unchanged reactant Na_2CO_3 and rest produced from NaHCO_3 .

The wt. of Na_2CO_3 produced by $\text{NaHCO}_3 = (2.752 - x)\text{g}$



$$(3.0 - x) \qquad (2.752 - x)$$

Applying POAC for Na atom

$$1 \times \text{moles of NaHCO}_3 = 2 \times \text{moles of Na}_2\text{CO}_3$$

$$\Rightarrow \frac{(3-x)}{84} = 2x \frac{(2.752-x)}{106}$$

$$\therefore x = 2.3244 \text{ g}$$

$$\therefore \% \text{ of Na}_2\text{CP}_3 = \frac{2.3244}{3} \times 100 = 77.48\%$$

Ex.30 10 g of a sample of a mixture of CaCl_2 and NaCl is treated to precipitate all the calcium as CaCO_3 . This CaCO_3 is heated to convert all the Ca to CaO and the final mass of CaO is 1.62 g. The percent by mass of CaCl_2 in the original mixture is:

- (A) 32.1 % (B) 16.2 % (C) 21.8 % (D) 11.0 %

Ans. (A)

Sol. $\text{CaCl}_2 + \text{NaCl} = 10 \text{ g}$

Let weight of $\text{CaCl}_2 = x \text{ g}$



1 mol 1 mol 1 mol

$$\frac{x}{111} \text{ mol} \quad \frac{x}{111} \text{ mol} \quad \frac{x}{111} \text{ mol}$$

$$\text{Mole of CaO} = \frac{1.62}{56}$$

$$\therefore \frac{x}{111} = \frac{1.62}{56} \Rightarrow x = 3.21 \text{ g}$$

$$\% \text{ of CaCl}_2 = \frac{3.21}{10} \times 100 = 32.1 \%$$

Do yourself-4

- 3.0 g an impure sample of sodium sulphate dissolved in water was treated with excess of barium chloride solution when 1.74 g of BaSO_4 was obtained as dry precipitate. Calculate the percentage purity of sample.
- If the percentage yield of given reaction is 30%, how many total moles of the gases will be produced, if 8 moles of NaNO_3 are taken initially:

$$\text{NaNO}_3(\text{s}) \longrightarrow \text{Na}_2\text{O}(\text{s}) + \text{N}_2(\text{g}) + \text{O}_2(\text{g}) \quad (\text{unbalanced})$$

(A) 4.2 mole (B) 2.4 mole (C) 4.8 mole (D) 2.1 mole
- A 5 g mixture of SO_2 and O_2 gases is reacted to form SO_3 gas. What should be the mass ratio of SO_2 and O_2 gases in mixture to obtain maximum amount of SO_3 gas:

(A) 4 : 1 (B) 3 : 2 (C) 2 : 3 (D) 1 : 4
- 25.4 g of iodine and 14.2g of chlorine are made to react completely to yield a mixture of ICl and ICl_3 . Calculate the number of moles of ICl and ICl_3 formed:

(A) 0.1 mole, 0.1 mole (B) 0.1 mole, 0.2 mole
 (C) 0.5 mole, 0.5 mole (D) 0.2 mole, 0.2 mole
- When 1 mole of A reacts with $\frac{1}{2}$ mole of B_2 ($\text{A} + \frac{1}{2} \text{B}_2 \longrightarrow \text{AB}$), 100 Kcal heat is liberated and when 1 mole of A reacted with 2 mole of B_2 ($\text{A} + 2\text{B}_2 \longrightarrow \text{AB}_4$), 200 Kcal heat is liberated. When 1 mole of A is completely reacted with excess, of B_2 to form AB as well as AB_4 , 140 Kcal heat is liberated calculate the mole of B_2 used.
 [Write your answer as number of mole of B_2 used $\times 10$]
- A solid mixture weighing 5.00 g containing lead nitrate and sodium nitrate was heated below 600°C until the mass of the residue was constant. If the loss of mass is 30 %, find the mass of lead nitrate and sodium nitrate in mixture.

$$2\text{Pb}(\text{NO}_3)_2 \xrightarrow{\Delta} 2\text{PbO} + 4\text{NO}_2\uparrow + \text{O}_2\uparrow$$

$$2\text{NaNO}_3 \xrightarrow{\Delta} 2\text{NaNO}_2 + \text{O}_2\uparrow$$

(At. wt. of Pb = 207, Na = 23, N = 14, O = 16)

5. Density:

It is of two type.

1. Absolute density
2. Relative density

(a) For liquid and solids:

$$\text{Absolute density} = \frac{\text{mass}}{\text{volume}}$$

$$\text{Relative density or specific gravity} = \frac{\text{density of the substance}}{\text{density of water at } 4^{\circ}\text{C}}$$

(b) For gasses:

$$\text{Absolute density (mass / volume)} = \frac{PM}{RT}$$

where, P is pressure of gas, M = mol. wt. of gas, R is the gas constant, T is the temperature.

Relative density and Vapour density:

Vapour density is defined as the density of the gas with respect to hydrogen gas at the same temperature and pressure.

$$\text{Vapour density} = \frac{d_{\text{gas}}}{d_{\text{H}_2}} = \frac{PM_{\text{gas}} / RT}{PM_{\text{H}_2} / RT}$$

$$\text{V.D.} = \frac{M_{\text{gas}}}{M_{\text{H}_2}} = \frac{M_{\text{gas}}}{2} \Rightarrow \boxed{M_{\text{gas}} = 2\text{V.D.}}$$

Ex.31 Find the relative density of SO_3 gas with respect to methane:

- (A) 8 (B) 3.5 (C) 2.5 (D) 5

Ans. (D)

$$\text{Sol. R.D.} = \frac{M_{\text{SO}_3}}{M_{\text{CH}_4}} = \frac{80}{16} = 5$$

Ex.32 The atomic mass of a metal is 27 u. If its valency is 3, the vapour density of the volatile metal chloride will be:

- (A) 66.75 (B) 321 (C) 267 (D) 80.25

Ans. (A)

Sol. Element must be Al

$$\text{Hence, volatile chloride will be } \text{AlCl}_3 \text{ so V.D.} = \frac{M_{\text{AlCl}_3}}{2} = \frac{133.5}{2} = 66.75$$

Ex.33 The density of water at 4°C is $1 \times 10^3 \text{ kg m}^{-3}$. Assuming no empty space to be present between water molecules, the volume occupied by one molecule of water is approximately:

- (A) $3 \times 10^{-23} \text{ mL}$ (B) $6 \times 10^{-23} \text{ mL}$ (C) $3 \times 10^{-22} \text{ mL}$ (D) $6 \times 10^{-22} \text{ mL}$

Ans. (A)

$$\begin{aligned} \text{Sol. } 1 \times 10^3 \text{ kg/m}^3 &= 1 \text{ g/mL.} \quad [\text{Since, } 1\text{m}^3 = 10^6 \text{ cm}^3 = 10^6 \text{ mL}]. \\ &= 1 \text{ gm/cc} \\ 6.022 \times 10^{23} \text{ H}_2\text{O molecule weigh } &\dots 18 \text{ g} \end{aligned}$$

$$1 \text{ H}_2\text{O molecule weight} \frac{18}{6.022 \times 10^{23}} \text{ g} = 3 \times 10^{-23} \text{ g}$$

$$d = \frac{\text{mass}}{\text{volume}}$$

$$\text{So, volume} = \frac{3 \times 10^{-23} \text{ g}}{1 \text{ (g / mL)}} = 3 \times 10^{-23} \text{ mL}$$

6. Average/Mean Atomic Mass:

The weighted average of the isotopic masses of the element's naturally occurring isotopes.

$$\text{Mathematically, average atomic mass of X (A}_x\text{)} = \frac{a_1x_1 + a_2x_2 + \dots + a_nx_n}{100}$$

Where:

$a_1, a_2, a_3 \dots$ atomic mass of isotopes.

and $x_1, x_2, x_3 \dots$ mole % of isotopes.

Ex.34 Naturally occurring chlorine is 75.53% Cl^{35} which has an atomic mass of 34.969 amu and 24.47% Cl^{37} which has a mass of 36.966 amu. Calculate the average atomic mass of chlorine:

- (A) 35.5 amu (B) 36.5 amu (C) 71 amu (D) 72 amu

Sol. (A)

Average atomic mass = (% of I isotope $\times$ Its atoms mass) + (% II isotope $\times$ its atomic mass)/100

$$= \frac{75.53 \times 34.969 + 24.47 \times 36.96}{100}$$

$$= 35.5 \text{ amu.}$$

Average Molar Mass or Average Gram Molecular Mass:

$$\text{The average molar mass of the different substance present in the container} = \frac{n_1M_1 + n_2M_2 + \dots + n_nM_n}{n_1 + n_2 + \dots + n_n}$$

Where:

$M_1, M_2, M_3 \dots$ are molar masses.

$n_1, n_2, n_3 \dots$ moles of substances.

Average molecule wt. = $\frac{\sum n_i M_i}{\sum n_i}$ where n_i = no. of moles of compound, m_i = molecular mass of compound

Ex.35 The molar composition of polluted air is as follows:

Gas	mole percentage composition
Oxygen	16%
Nitrogen	80%
Carbon dioxide	03%
Sulphurdioxide	01%

What is the average molecular weight of the given polluted air?

Ans. 29.48

$$\text{Sol. } M_{\text{avg}} = \frac{16 \times 32 + 80 \times 28 + 44 \times 3 + 64 \times 1}{100} = \frac{512 + 2240 + 132 + 64}{100} = \frac{2948}{100} = 29.48$$

Degree of Dissociation (α):

Degree of dissociation represents the fraction of one mole dissociated into the products.

(Defined for one mole of substance)

So, α = no. of moles dissociated / initial no. of moles taken

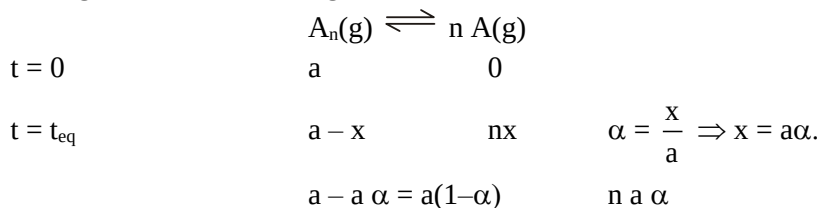
= fraction of moles dissociated out of 1 mole.

Note: % dissociation = $\alpha \times 100$

Suppose 5 moles of PCl_5 is taken and if 2 moles of PCl_5 dissociated then $\alpha = \frac{2}{5} = 0.4$

Relationship Between Average Molar Mass & Degree of Dissociation (α):

Let a gas A_n dissociates to give n moles of A as follows:



Total no. of moles = $a - a\alpha + n a \alpha = [1 + (n-1)\alpha] a$

Average molecular weight of mixture(g) = $\frac{\text{molecular weight of } A_n(g)}{\text{total no. of moles at equilibrium}} = \frac{a.M_{th}}{a(1 + (n-1)\alpha)}$

$$M_{avg.} = \frac{M_{th}}{[1 + (n-1)\alpha]}$$

where, M_{th} = theoretical molecular weight (n = atomicity)

$$M_{mixture} = \frac{M_{A_n}}{[1 + (n-1)\alpha]}, \quad M_{A_n} = \text{Molar mass of gas } A_n$$

Vapour density (V.D):

Density of the gas divided by density of hydrogen under same temperature & pressure is called vapour density.

$$D = \text{vapour density without dissociation} = \frac{M_{A_n}}{2}$$

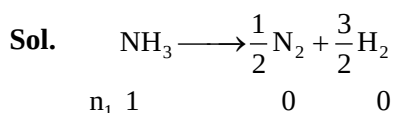
d = vapour density of mixture = average vapour density

$$\frac{D}{d} = 1 + (n-1)\alpha$$

$$\alpha = \frac{D-d}{(n-1)d} = \frac{M_T - M_o}{(n-1)M_o}$$

Ex.36 NH_3 decomposes into N_2 & H_2 . If average molar mass of reaction mixture is 10 then, find α ?

Ans. 0.7



$$n_1 \quad 1 \quad \quad 0 \quad \quad 0$$

$$1-\alpha \quad \frac{\alpha}{2} \quad \quad \frac{3\alpha}{2}$$

$$10 = \frac{17}{1-\alpha + \frac{\alpha}{2} + \frac{3\alpha}{2}} \Rightarrow 10 = \frac{17}{1+\alpha} \Rightarrow 1+\alpha = 1.7 \Rightarrow \alpha = 0.7$$

7. Percentage Composition and Molecular Formula:

Here we are going to find out the percentage of each element in the compound by knowing the molecular formula of compound.

We know that according to law of definite proportion any sample of a pure compound always possess constant ratio with their combining elements.

Example:

Every molecule of ammonia always has formula NH_3 irrespective of method of preparation or sources. i.e. 1 mole of ammonia always contains 1 mol of N and 3 mole of H. In other words 17 g of NH_3 always contains 14 g of N and 3 g of H. Now find out % of each element in the compound.

$$\text{Mass \% of N in NH}_3 = \frac{\text{Mass of N in 1 mol NH}_3}{\text{Mass of 1 mol of NH}_3} \times 100 = \frac{14\text{g}}{17} \times 100 = 82.35\%$$

$$\text{Mass \% of H in NH}_3 = \frac{\text{Mass of H in 1 mol NH}_3}{\text{Mass of 1 mol of NH}_3} \times 100 = \frac{3}{17} \times 100 = 17.65\%$$

Empirical and Molecular Formula:

We have just seen that knowing the molecular formula of the compound we can calculate percentage composition of the elements. Conversely if we know the percentage composition of the elements initially, we can calculate the relative number of atoms of each element in the molecules of the compound. This gives us the empirical formula of the compound. Further if the molecular mass is known then the molecular formula can be easily determined.

Thus, the empirical formula of a compound is a chemical formula showing the relative number of atoms in the simplest ratio, the molecular formula gives the actual number of atoms of each element in a molecule.

Empirical formula: An empirical formula represents the simple whole number ratio of various atoms present in a compound,

Molecular formula: whereas, the molecular formula shows the exact number of different types of atoms present in a molecule of a compound

The molecular formula is generally an integral multiple of the empirical formula.

That is: $\boxed{\text{molecular formula} = \text{empirical formula} \times n}$ where, $n = \frac{\text{molecular formula mass}}{\text{empirical formula mass}}$

Ex.37 An organic substance containing carbon, hydrogen and oxygen gave the following percentage composition.

C = 40.687 %; H = 5.085 % and O = 54.228 %

The molecular weight of the compound is 118. Calculate the molecular formula of the compound.

Ans. $\text{C}_4\text{H}_6\text{O}_4$

Sol. Step -1

To calculate the empirical formula of the compound.

Element	Symbol	Percentage of element	At. mass of element	Relative no. of atoms = $\frac{\text{Percentage}}{\text{At.mass}}$	Simplest Atomic ratio	Simplest whole no. atomic ratio
Carbon	C	40.687	12	$\frac{40.687}{12} = 3.390$	$\frac{3.390}{3.389} = 1$	2
Hydrogen	H	5.085	1	$\frac{5.085}{1} = 5.035$	$\frac{5.085}{3.389} = 1.5$	3
Oxygen	O	54.228	16	$\frac{54.228}{16} = 3.389$	$\frac{3.389}{3.389} = 1$	2

Step - 2

To calculate the empirical formula mass.

The empirical formula of the compound is $C_2H_3O_2$.

$\therefore$ Empirical formula mass

$$= (2 \times 12) + (3 \times 1) + (2 \times 16) = 59.$$

Step - 3

To calculate the value of 'n'

$$n = \frac{\text{molecular mass}}{\text{Empirical formula mass}} = \frac{118}{59} = 2$$

Step - 4

To calculate the molecular formula of the salt

Molecular formula = $n \times (\text{Empirical formula})$

$$= 2 \times C_2H_3O_2 = C_4H_6O_4$$

Thus, the molecular formula is $C_4H_6O_4$.

Ex.38 Chlorophyll the green colouring material of plants contains 3.68 % of magnesium by mass. Calculate the number of magnesium atom in 5.00 g of the complex.

Sol. Mass of magnesium in 5.0 g of complex = $\frac{3.68}{100} \times 5.00 = 0.184 \text{ g}$

Atomic mass of magnesium = 24

24 g of magnesium contain = 6.023×10^{23} atoms

$$0.184 \text{ g of magnesium would contain} = \frac{6.023 \times 10^{23}}{24} \times 0.184 = 4.617 \times 10^{21} \text{ atom}$$

Therefore, 5.00 g of the given complex would contain 4.617×10^{21} atoms of magnesium.

Ex.39 A sample of $CaCO_3$ has Ca = 40%, C = 12% and O = 48% by mass. If the law of constant proportions is true, then the mass of Ca in 5 g of $CaCO_3$ obtained from another source will be:

- (A) 0.2 g (B) 2 g (C) 0.6 g (D) Cannot be determined

Ans. (B)

Sol. Mass of Ca = $5 \times \frac{40}{100} = 2 \text{ g}$.

8. Experimental Methods to Determine Atomic & Molecular Masses:

For determination of atomic mass:

(a) Dulong's & Pettit's law:

In case of metals, it is observed that product of atomic weight and specific heat capacity is constant. The equation connecting the two parameters was given by Dulong's and Petit's Law.

Atomic weight of metal $\times$ specific heat capacity (cal/gm°C) = 6.4.

It should be remembered that this law is an empirical observation and this gives an approximate value of atomic weight. Also, this law can be applied only to metals at high temperature conditions only.

Experimental methods for molecular mass determination:

- (a) Victor Meyer's Method
- (b) Silver Salt Method
- (c) Chloroplatinate Salt Method

(a) Victor Meyer's Method: (Applicable for volatile substance):

A known mass of the volatile substance taken in the Hoffmann's bottle and is vapourised by throwing the Hoffmann's bottle into the Victor Meyer's tube. The vapour displaces an equal volume of the moist air. Which vapours is measured at the room temperature and atmospheric pressure. The barometric pressure and the room temperature is recorded. Following diagram gives the experimental set-up for the Victor-Meyer's process.

Calculation involved

Let the mass of the substance taken by = W g

Volume of moist vapours collected = V cm³

Room temperature = T K

Barometric pressure = P mm

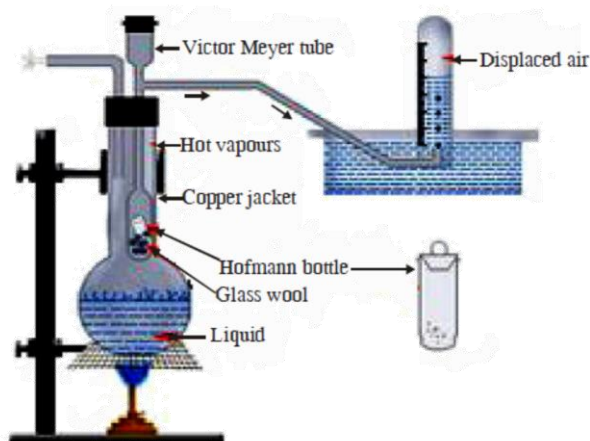
Aqueous tension at T K = p mm

Pressure of dry vapour = $(P - p)$ mm

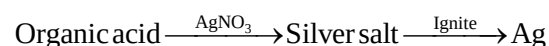
Calculation of molecular mass (M)

$$\frac{(P - p)}{760} \times \frac{V}{1000} = \frac{w}{M} \times RT$$

$$\Rightarrow M = \frac{w \times RT \times 760 \times 1000}{(P - p) \times V}$$

**(b) Silver salt Method: (A used for organic acids):**

A known mass of the acid is dissolved in water followed by the subsequent addition of silver nitrate solution till the precipitation of silver salt is complete. The precipitate is separated, dried, weighed



and ignited till decomposition is complete. The residue of pure silver left behind is weighed.

Calculations involved

Let the mass of the silver salt formed = W g

The mass of Ag formed = x g

Let us understand to calculations by considering the monobasic acid MX .



Mass of the salt that gives x gm of Ag = W g

$$\text{Mass of the salt gives 108g (1g-atom) of Ag} = \frac{108W}{x} \text{ g}$$

$$\text{Molar mass of salt} = \frac{108W}{x} \text{ g}$$

Molar mass of acid = (Molar mass of salt) – (Atomic mass of Ag) + (Atomic mass of H)

$$= \frac{108W}{x} - 108 + 1 = \left(\frac{108W}{x} - 107 \right) \text{ g mol}^{-1}$$

For polybasic acid of the type H_nX (n is basicity)



Mass of the silver that gives x g of Ag = W g

$$\text{Mass of the silver that gives (108n g) of Ag} = \frac{108nW}{x} \text{ g}$$

$$\text{Molar mass of salt} = \frac{108 \times nW}{x} \text{ g}$$

$$\text{Molar mass of acid} = (\text{molar mass of salt}) = \frac{108 \times nW}{x} - n \times 108 - n \times 1$$

$$= n \left(\frac{108W}{x} - 107 \right) \text{ g mol}^{-1}$$

(c) Platinic chloride Method: (Applicable for finding the molecular masses of organic bases)

A known mass of organic base is allowed to react with chloroplatinic acid (H_2PtCl_6) in conc. HCl to form insoluble platinic chloride. The precipitate of platinic chloride is separated, dried, weighed and subsequently ignited till decomposition is complete. The residue left is platinum which is again weighed. The molecular mass is then calculated by knowing the mass of the platinic chloride salt and that of platinum left.

If B represents the molecule of monoacidic organic base, then the formula of platinic chloride salt is $\text{B}_2\text{H}_2\text{PtCl}_6$



Let the mass of platinic chloride salt = W g, The mass of the platinum residue left = x g

It may be noted that salt formed with diacidic base would be $\text{B}_2(\text{H}_2\text{PtCl}_6)_2$

With triacidic base it would be $\text{B}_2(\text{H}_2\text{PtCl}_6)_3$ and with polyacidic base would be $\text{B}_2(\text{H}_2\text{PtCl}_6)_n$.

$$\text{Mass of salt which gives 195 g (1 g-atom) of Pt} = \frac{W \times 195}{x} \text{ g}$$

$$\text{Molar mass of salt} = \frac{W \times 195}{x} \text{ g mol}^{-1}$$

Now from the formula $\text{B}_2(\text{H}_2\text{PtCl}_6)$

Molar mass of salt = (2 × molar mass of base) + (Molar mass of H_2PtCl_6)

$$\text{Molar mass of base} = \frac{1}{2} (\text{molar mass of salt} - \text{Molar mass of } \text{H}_2\text{PtCl}_6)$$

$$= \frac{1}{2} \left(\frac{W \times 195 \times n}{x} - n \times 410 \right) = \frac{n}{2} \left(\frac{w \times 195}{x} - 410 \right) \text{ g mol}^{-1}$$

Do yourself-5

- 120 g Mg is burnt in air to give a mixture of MgO and Mg_3N_2 . The mixture is now dissolved in HCl to form MgCl_2 and NH_4Cl . If 107 g NH_4Cl is produced, then determine the moles of MgCl_2 formed:
(A) 2.5 (B) 4 (C) 2 (D) 5
- Penicillin V was treated chemically to convert the sulphur present to barium sulphate, BaSO_4 . A 9.6 mg sample of penicillin V gave 4.66 mg BaSO_4 . The percentage of sulphur in Penicillin V is x %. If there is one sulphur atom in the molecule, the molecular weight of Pencillin V is y amu.
Report your answer as y/x.
- From the following reaction sequence:

$$\text{Cl}_2 + 2\text{KOH} \longrightarrow \text{KCl} + \text{KClO} + \text{H}_2\text{O}$$

$$3\text{KClO} \longrightarrow 2\text{KCl} + \text{KClO}_3$$

$$4\text{KClO}_3 \longrightarrow 3\text{KClO}_4 + \text{KCl}$$
 Calculate the mass of chlorine needed to produce 138.5 g of KClO_4 :
 (A) 142 g (B) 284 g (C) 432 g (D) None of these
- The density of air at STP is 0.0013 g mL^{-1} . Its vapour density is:
 (A) 0.015 (B) 15 (C) 1.5 (D) Data insufficient
- $\text{SO}_3(\text{g}) \rightleftharpoons \text{SO}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g})$
 If observed vapour density of mixture at equilibrium is 35 then find out value of α :
 (A) 0.28 (B) 0.38 (C) 0.48 (D) 0.58
- A sample of a compound contains 9.75 g Zn, 9×10^{22} atoms of Cr and 0.6 gram-atoms of O. What is empirical formula of compound? (Atomic Mass Zn = 65)
 (A) ZnCrO_4 (B) ZnCr_2O_4 (C) Zn_2CrO_4 (D) None of these
- An organic compound on analysis was found to contain 0.032% of sulphur by mass. The molecular mass of the compound, if it's one molecule contains two sulphur atoms, is:
 (A) 100000 u (B) 10000 u (C) 20000 u (D) 200000 u
- In an organic compound of molar mass 108 g mol^{-1} C, H and N atoms are present in 9 : 1 : 3.5 by weight. Molecular formula can be:
 (A) $\text{C}_6\text{H}_8\text{N}_2$ (B) $\text{C}_7\text{H}_{10}\text{N}$ (C) $\text{C}_5\text{H}_6\text{N}_3$ (D) $\text{C}_4\text{H}_{18}\text{N}_3$
- At 100°C and 1 atmp, if the density of liquid water is 1.0 g cm^{-3} and that of water vapour is 0.0006 g cm^{-3} , then the volume occupied by water molecules in 1 L of steam at that temperature is:
 (A) 6 cm^3 (B) 60 cm^3 (C) 0.6 cm^3 (D) 0.06 cm^3

Solutions:

A mixture of two or more substances can be a solution. We can also say that a solution is a homogeneous mixture of two or more substances 'Homogeneous' means 'uniform throughout'. Thus a homogeneous mixture, i.e., a solution, will have uniform composition throughout.

9. Concentration Terms:

The following concentration terms are used to express the concentration of a solution. These are:

1. Molarity (M) 2. Molality (m) 3. Mole fraction (x)
4. % calculation 5. ppm

* Remember that all of these concentration terms are related to one another. By knowing one concentration term you can also find the other concentration terms. Let us discuss all of them one by one.

Molarity (M):

The number of moles of a solute dissolved in 1 L (1000 ml) of the solution is known as the molarity of the solution.

That is, Molarity of solution = $\frac{\text{number of moles}}{\text{volume of solution in litre}}$

Let a solution is prepared by dissolving w g of solute of mol. wt. M in V mL water.

$$\therefore \text{Number of moles of solute dissolved} = \frac{w}{M}$$

$$\therefore V \text{ mL water have } \frac{w}{M} \text{ mole of solute}$$

$$\therefore 1000 \text{ mL water have } \frac{w \times 1000}{M \times V(\text{in mL})}$$

$$\Rightarrow \therefore \text{Molarity (M)} = \frac{w \times 1000}{(\text{Mol. wt of solute}) \times V(\text{in mL})}$$

Some other relations may also useful.

$$\text{Number of millimoles} = \frac{\text{mass of solute}}{(\text{Mol. wt. of solute})} \times 1000 = (\text{Molarity of solution} \times V_{\text{ml}})$$

$$\text{Molarity of solution may be also given as: } \frac{\text{Number of millimole of solute}}{\text{Total volume of solution in ml}}$$

Molarity is a unit that depends upon temperature. It decreases as temperature increases.

Ex.40 Find the mass of solute and solvent in 100 mL, 1 M NaOH solution having density 1.5 g/mL.

- (A) 40 g, 110 g (B) 4 g, 150 g (C) 4 g, 146 g (D) 40 g, 150 g

Ans. (C)

Sol. Mole of NaOH = molarity $\times$ volume (l) = $1 \times 0.1 = 0.1$

Mass of NaOH = $0.1 \times 40 = 4\text{gm}$

Mass of solution = volume $\times$ density = $100 \times 1.5 = 150\text{gm}$

Hence: mass of solvent = $150 - 4 = 146 \text{ g}$.

Ex.41 Molality of pure water if its density is 0.936 gm/ml

- (A) 50 (B) 55.56 (C) 57.56 (D) 56.56

Ans. (B)

$$\text{Sol. } m = \frac{1000}{\text{M.W.}} = \frac{1000}{18} = 55.56$$

Molality (m):

The number of moles of solute dissolved in 1000 g (1 kg) of a solvent is known as the molality of the solution.

$$\text{That is, molality} = \frac{\text{number of moles of solute}}{\text{mass of solvent in gram}} \times 100$$

Let y g of a solute is dissolved in x g of a solvent. The molecular mass of the solute is m. Then y/m mole of the solute are dissolved in x g of the solvent. Hence,

$$\text{Molality} = \frac{y}{m \times x} \times 1000$$

Ex.42 225 gm of an aqueous solution contains 5 gm of urea. What is the concentration of the solution in terms of molality. (Mol. wt. of urea = 60)

Ans. 0.332.

Sol. Mass of urea = 5 gm

Molecular mass of urea = 60

$$\text{Number of moles of urea} = \frac{5}{60} = 0.083$$

Mass of solvent = (225 – 5) = 220 gm

$$\therefore \text{Molality of the solution} = \frac{\text{Number of moles of solute}}{\text{Mass of solvent in gram}} \times 1000$$

$$= \frac{0.083}{220} \times 1000 = 0.377$$

Ex.43 A solution is made by dissolving CaBr₂ in water (solvent) such that mass fraction of solute and solvent is same in the solution. The molality of solution is:

- (A) 2.5 m (B) 55.55 m (C) 2 m (D) 5 m

Ans. (D)

$$\text{Sol. } m = \frac{w / 200 \times 1000}{w} = 5$$

Mole fraction (x):

The ratio of number of moles of the solute or solvent present in the solution and the total number of moles present in the solution is known as the mole fraction of substances concerned.

Let number of moles of solute in solution = n

Number of moles of solvent in solution = N

$$\therefore \text{Mole fraction of solute (x}_1\text{)} = \frac{n}{n + N}$$

$$\therefore \text{Mole fraction of solvent (x}_2\text{)} = \frac{N}{n + N} \Rightarrow \text{also } x_1 + x_2 = 1$$

% Calculation: The concentration of a solution may also express in terms of percentage in the following ways.

(i) % weight by weight (% w/w): It is given as mass of solute present in per 100 g of solution.

$$\text{i.e. \% w/w} = \frac{\text{mass of solute in g}}{\text{mass of solution in g}} \times 100$$

[X % by mass means 100 g solution contains X g solute;

$\therefore$ (100 – X) g solvent]

(ii) **% weight by volume (% w/V):** It is given as mass of solute present in per 100 mL of solution.

$$\text{i.e. } \% \text{ w/v} = \frac{\text{mass of solute in g}}{\text{volume of solution in mL}} \times 100$$

$\left[X \% \left(\frac{w}{V}\right)\right]$ means 100 mL solution contains X g solute]

(iii) **% volume by volume (% V/V):** It is given as volume of solute present in per 100 mL solution.

$$\text{i.e. } \% \text{ V/V} = \frac{\text{Volume of solute}}{\text{Volume of solution in mL}} \times 100$$

Ex.44 0.25 g of a substance is dissolved in 6.25 g of a solvent. Calculate the percentage amount of the substance in the solution.

Ans. 3.8%.

Sol. wt. of solution = 0.25 + 6.25 = 6.50.

$$\text{so } \% \text{ (w/w)} = \frac{0.25}{6.50} \times 100 = 3.8\%.$$

Ex.45 0.5 g of a substance is dissolved in 25 g of a solvent. Calculate the percentage amount of the substance in the solution.

Ans. 1.96

Sol. Mass of substance = 0.5 g; Mass of solvent = 25 g

$$\therefore \text{Percentage of the substance (w/w)} = \frac{0.5}{0.5 + 25} \times 100 = 1.96$$

$$\text{Parts per million (ppm): } \frac{\text{Mass of solute}}{\text{Mass of solvent}} \times 10^6 \cong \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 10^6$$

Do yourself-6

- Calculate the molarity when:
 - 4.9 g H_2SO_4 acid dissolved in water to result 500 mL solution.
 - 2 gram-molecules of KOH dissolved in water to result 500 mL solution.

(A) (a) 0.1 M, (b) 0.07 M (B) (a) 0.4 M, (b) 4 M
(C) (a) 0.4 M, (b) 0.07 M (D) (a) 0.1 M, (b) 4 M
- Calculate the volume in litre of 0.1 M solution of HCl which contains 0.365 g HCl?

(A) 10^{-2} L (B) 0.1 L (C) 1 L (D) 10 L
- What volume of a 0.8 M solution contains 100 millimoles of the solute?

(A) 80 mL (B) 125 mL (C) 125 L (D) 80 L
- Which of the following methods of expressing concentration of a solution is/are independent of temperature?

(A) Molality (B) % w/w
(C) Mole fraction of solute (D) All of these
- 20 cm^3 of an alcohol is dissolved in 80 cm^3 of water. Calculate the percentage of alcohol in solution.
- Calculate the amount of 75% pure NaI required to prepare 5 litre of 0.5 M solution.

(A) 281.25 g (B) 500 g (C) 923.33 g (D) 519.375 g

10. Dilution and Intermixing of Solutions:

Dilution: Whenever a given solution of known concentration i.e. normality and molarity (known as standard solution) is diluted (adding solvent), the number of millimoles (or milli equivalents) of solute remain unchanged. The concentration of solution however changes. In such a case if:

M_1 = Molarity of original solution; V_1 = volume of original solution and

M_2 = normality of diluted solution; V_2 = total volume of diluted solution

Since the number of millimoles remains same,

$$\Rightarrow M_1V_1 = M_2V_2$$

Ex.46 Calculate the resultant molarity of following:

(A) 200 ml 1M HCl + 300 ml water

(B) 1500 ml 1M HCl + 18.25 g HCl

(C) 200 ml 1M HCl + 100 ml 0.5 M H_2SO_4

(D) 200 ml 1M HCl + 100 ml 0.5 M HCl

Ans. (A) 0.4 M

(B) 1.33 M

(C) 1 M

(D) 0.83 M.

Sol. (A) Final molarity = $\frac{200 \times 1 + 0}{200 + 300} = 0.4$ M.

(B) Final molarity = $\frac{1500 \times 1 + \frac{18.25 \times 1000}{36.5}}{1500} = 1.33$ M

(C) Final molarity of H^+ = $\frac{200 \times 1 + 100 \times 0.5 \times 2}{200 + 100} = 1$ M.

(D) Final molarity = $\frac{200 \times 1 + 100 \times 0.5}{200 + 100} = 0.83$ M.

Ex.47 The molarity of Cl^- in an aqueous solution which was (w/V) 2% NaCl, 4% $CaCl_2$ and 6% NH_4Cl will be:

(A) 0.342

(B) 0.721

(C) 1.12

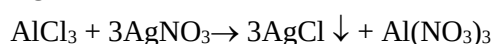
(D) 2.18

Ans. (D)

Sol. Moles of Cl^- in 100 ml of solution = $\frac{2}{58.5} + \frac{4}{111} \times 2 + \frac{6}{53.5} = 0.2184$

Molarity of Cl^- = $\frac{0.2184}{100} \times 1000 = 2.184$.

Ex.48 How many milli-litres of 0.2 M $AlCl_3$ solution is required to precipitate all the Ag^+ from 45 ml of a 0.2 M $AgNO_3$ solution:



(A) 15 ml

(B) 30 ml

(C) 45 ml

(D) 60 ml

Ans. (A)

Sol. $AlCl_3 + 3AgNO_3 \rightarrow 3AgCl \downarrow + Al(NO_3)_3$

45 × 0.2 milli moles

$$\frac{1}{3} \times 45 \times 0.2 \text{ milli mol}$$

$$\frac{1}{3} \times 0.45 \times 0.2 = 0.2 \times V$$

$$\Rightarrow V = 15 \text{ ml}$$

Ex.49 The specific gravity of a solution is 1.8, having 62% by weight of acid. It is to be diluted to specific gravity of 1.2. What volume of water should be added to 100 ml of this solution?

Ans. 300 mL

Sol. Let, to 100 ml of given acid solution (sp. gr 1.8) x ml. of water is added.

$$\therefore \text{the total volume of resulting solution} = (100 + x) \text{ ml}$$

$$\therefore \text{the total weight of resulting solution} = (100 + x) \times 1.2 \text{ gm.}$$

$$\text{weight of acid present in the given acid solution (per 100 ml)} = 100 \times 1.8 \times 0.62$$

$$\therefore \text{the amount of water present in 100 ml of given acid solution} = 1.8 \times 100 \times 0.38$$

$$\therefore \text{total wt of acid present in the diluted solution} = (100 + x) 1.2 - x - 180 \times 0.38 = 1.8 \times 100 \times 0.62$$

$$\therefore 120 + 0.2x = 180 \quad \text{or} \quad x = 300$$

$$\therefore \text{To lower sp. gravity of the given acid solution to 1.2, we are to add 300 ml of water per 100 ml of acid solution (sp gr. 1.2).}$$

Ex.50 How would you prepare exactly 3 L of 1 M NaOH solution by mixing proportions of stock solutions of 2.5 M NaOH and 0.4 M NaOH, if no water is to be used? Find the ratio of the volume (V_1/V_2).

(A) 1 : 3

(B) 3 : 7

(C) 2 : 5

(D) Data insufficient

Ans. (C)

Sol. $M_1V_1 + M_2V_2 = M_TV_T$

$$2.5 V_1 + 0.4 V_2 = 3 \times 1$$

$$2.5 V_1 + 0.4 (3 - V_1) = 3 \Rightarrow 2.5 V_1 + 1.2 - 0.4 V_1 = 3$$

$$2.1 V_1 = 1.8$$

$$V_1 = \frac{1.8}{2.1} = \frac{6}{7}$$

$$V_2 = 3 - \frac{6}{7} = \frac{15}{7}$$

$$\frac{V_1}{V_2} = \frac{6}{7} \times \frac{7}{15} = 2 : 5$$

11. Interconversion of Concentration Terms:

Comprehension # (Q.51 to Q.52)

	Molarity(mol/L)	Molality(mol/Kg)	Density (g/mL)	Gram molecular mass of solute
Solution-1	a	-	d_1	P
Solution-2	-	b	d_2	Q
Solution-3	1	-	1.060	60

Now answer the following questions:

Ex.51 What is molality of solution-1:

(A) $\frac{(1000 \times a)}{(1000 \times d_1) - aP}$

(B) $\frac{1000 d_1}{1000 a - P}$

(C) $\frac{a}{1000 d_1 - aP}$

(D) None of these

Ans. (A)

Sol. For solution 1 'a' mole of solute are present in 1000 ml of solution.

$$\text{wt. of solution} = 1000 \times d_1 \text{ g}$$

$$\text{wt. of solute} = aP \text{ g}$$

$$\text{So, Molality} = \left[\frac{a \times 1000}{1000 \times d_1 - aP} \right]$$

Ex.52 What is the molarity of solution-2:

(A) $\frac{b \times d_2}{1000 + bQ}$

(B) $\frac{b \times 1000 \times d_2}{1000 + bQ}$

(C) $\frac{1000 \times bQ}{1000 + bd_2}$

(D) None of these

Ans. (B)

Sol. For solution 2 'b' moles of solute are present in 1000 g of solvent.

wt. of solution = 1000 + bQ

vol. of solution = $\frac{1000 + bQ}{d_2}$

Molality = $\frac{b \times 1000}{1000 + bQ}$

= $\frac{b \times 1000 \times d_2}{1000 + bQ}$

Ex.53 The molarity of the solution containing 2.8% (mass/volume) solution of KOH is:

(Given atomic mass of K = 39) is:

(A) 0.1 M

(B) 0.5 M

(C) 0.2 M

(D) 1 M

Sol. Weight of KOH = 2.8 gram

Volume of solution = 100 ml

$M = \frac{2.8 \times 1000}{56 \times 100} = \frac{5}{49} = 0.5 \text{ M}$

Ex.54 What is the mole fraction of ethanol in 20% by weight solution in water?

(A) 0.095

(B) 0.089

(C) 0.9

(D) 1.2

Ans. (B)

Sol. 100 gm of solution contain 20 gm C_2H_5OH and 80 gm of water

$\therefore$ moles of ethanol present = $\frac{20}{46} = 0.435$

(mol. wt of ethanol = 46)

$\therefore$ moles of water present = $\frac{80}{18} = 4.444$

Total no. of moles = $0.435 + 4.444 = 4.879$

$\therefore$ mole fraction of C_2H_5OH

= $\frac{0.435}{4.879} = 0.089$

Do yourself-7

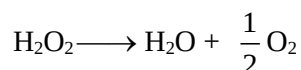
- What volume of water is required to make 0.2 M solution from 16 mL of a 0.5 M solution?
(A) 24 mL (B) 40 mL (C) 6.4 mL (D) 20 mL
- What approximate volume of 0.40 M Ba(OH)₂ solution must be added to 50.0 mL of 0.30 M NaOH solution to get a solution in which the molarity of the OH⁻ ions is 0.50 M?
(A) 33 mL (B) 66 mL (C) 133 mL (D) 100 mL
- 100 mL 30% (w/v) NaOH solution is mixed with 100 mL 90% (w/v) NaOH solution. The molarity of final solution is:
(A) 30M (B) 15M (C) 7.5M (D) 2M
- Molality (m) of a sulphuric acid solution in which the mol fraction of water is 0.85 is:
(A) 4.9 (B) 9.8 (C) 19.6 (D) Can't be determined
- The molality of a sulphuric acid solution is 0.2. Total weight of the solution having 100 g of solvent is about:
(A) 119.6 g (B) 109.8 g (C) 104.9 g (D) 102 g

12. Some Special Concentration Terms:**Volume Strength of H₂O₂:**

Strength of H₂O₂ is represented as 10V, 20 V, 30 V etc.

20V H₂O₂ means **one litre** of this sample of H₂O₂ on decomposition gives **20L of O₂** gas at **STP**.

Decomposition of H₂O₂ is given as :



$$1 \text{ mole} \quad \frac{1}{2} \times 22.7 \text{ L O}_2 \text{ at STP}$$

$$= 34\text{g} \quad = 11.35 \text{ L O}_2 \text{ at STP}$$

$$\text{Molarity of H}_2\text{O}_2 \text{ (M)} = \frac{\text{Volume strength of H}_2\text{O}_2}{11.35}$$

$$\text{Strength (in g/L)} = \text{Molarity} \times \text{Mol. Wt} = \text{Molarity} \times 34$$

Ex.55 A fresh H₂O₂ solution is labeled 11.35 V at STP. This solution has the same concentration as a solution which is:

- (A) 3.4% (w / w) (B) 3.4% (v / v) (C) 3.4% (w / v) (D) None of these

Ans. (C)

Sol. Molarity of $\text{H}_2\text{O}_2 = \frac{\text{vol. strength}}{11.2} = \frac{11.35}{11.35} = 1$

Now, $\%(\text{w/v}) = \frac{\text{wt. of solute in g}}{\text{wt. of solution in mL}} \times 100$

$= \text{Molarity} \times \text{Mol. wt. of solute} \times \frac{1}{10}$

$= 1 \times 34 \times \frac{1}{10} = 3.4\%$

Percentage Labeling of Oleum:

Oleum is SO_3 dissolved in 100% H_2SO_4 . Sometimes, oleum is reported as more than 100% by weight, say $y\%$ (where $y > 100$). This means that $(y - 100)$ grams of water, when added to 100 g of given oleum sample, will combine with all the free SO_3 in the oleum to give 100% sulphuric acid.

Hence, weight % of free SO_3 in oleum $= 80(y - 100)/18$

Ex.56 What volume of water is required (in mL) to prepare 1 L of 1 M solution of H_2SO_4 (density = 1.5g/mL) by using 109% oleum and water only (Take density of pure water = 1 g/mL).

Ans. 1410.09 mL

Sol. 1 mole H_2SO_4 in 1L solution = 98 g H_2SO_4 in 1500 g solution = 98 g H_2SO_4 in 1402 g water.

Also, in 109% oleum, 9 g H_2O is required to form 109 g pure H_2SO_4 & so, to prepare 98 g H_2SO_4 , water needed is $9/109 \times 98 = 8.09$ g.

Total water needed = 1402 + 8.09 = 1410.09 g = 1410.09 mL

Ex.57 A 50 gm oleum sample contains $\left(\frac{400}{49}\right)$ gm of combined SO_3 . Find percent label of the oleum sample.

Ans. (118)

Sol. Combined $\text{SO}_3 = \left(\frac{400}{49}\right)$ g is present in H_2SO_4

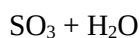
$$\text{mole of } \text{SO}_3 = \frac{\frac{400}{49}}{80} = \frac{5}{49}$$

$$\text{mole of } \text{H}_2\text{SO}_4 \text{ in oleum} = \frac{5}{49}$$

$$\text{In 50 g oleum mass of } \text{H}_2\text{SO}_4 = \frac{5}{49} \times 98 = 10 \text{ g}$$

$$100 \text{ g oleum mass of } \text{H}_2\text{SO}_4 = 20 \text{ g}$$

$$\text{mass of } \text{SO}_3 = 100 - 20 = 80 \text{ g}$$



$$\text{mass } 80 \text{ g}$$

$$\text{mol} = \frac{80}{80} = 1 \text{ mol} = 18 \text{ g}$$

$$\% \text{ labeling} = (100 + 18) \% = 118 \%$$

13. Eudiometry or Gas Mixture Analysis:

Gaseous reactions are carried out in a special type of tube known as eudiometer tube. The tube is graduated in millimeters for volume measurement. The reacting gases taken in the eudiometer tube are exploded by sparks. The volumes of the products of a gas are determined by absorbing them in suitable reagents,

Eg. Solvent	gas(es) absorb
KOH	CO ₂ , SO ₂ , Cl ₂
Ammonical Cu ₂ Cl ₂	CO
Turpentine oil	O ₃
Alkaline pyrogallol	O ₂
Water	NH ₃ , HCl
CuSO ₄	H ₂ O

Eudiometry is mainly based on Avogadro's law i.e. $V \propto n$ at the same temperature and pressure.

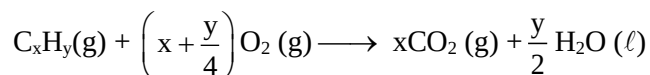
$\therefore$ The mole concept may be applied in solving the problems, keeping in mind that in a gaseous reaction the relative volumes (measured under identical conditions) of each reactant and product represent their relative numbers of moles.



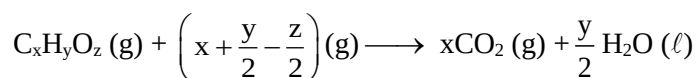
* Generally, explosions are carried out at STP and H₂O is assumed to be in liquid state, means its volume is negligible as compared to product gases.

Burning of hydrocarbon:

1. Hydrocarbon containing carbon and hydrogen only.



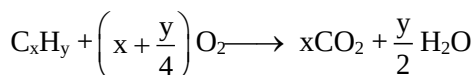
2. Hydrocarbon containing carbon and hydrogen and oxygen.



Ex.58A A gaseous hydrocarbon requires 6 times its own volume of O₂ for complete oxidation and produces 4 times its volume of CO₂. What is its formula?

Ans. C₄H₈

Sol. The balanced equation for combustion



$$1 \text{ volume} \left(x + \frac{y}{4}\right) \text{ volume}$$

$$\therefore x + \frac{y}{4} = 6 \text{ (by equation)}$$

$$\text{or } 4x + y = 24$$

.....(1)

Again $x = 4$ since evolved CO₂ is 4 times that of hydrocarbon

$$\therefore 16 + y = 24 \text{ or } y = 8 \therefore \text{ formula of hydrocarbon } C_4H_8$$

Ex.59 7.5 ml of a gaseous hydrocarbon was exploded with 36 ml of oxygen. The volume of gases on cooling was found to be 28.5 ml, 15 ml of which was absorbed by KOH and the rest was absorbed in a solution of alkaline pyrogallol. If all volumes are measured under same conditions, the formula of hydrocarbon is:

- (A) C_3H_4 (B) C_2H_4 (C) C_2H_6 (D) C_3H_6

Ans. (B)

Sol. $C_xH_y + \left(x + \frac{y}{4}\right) O_2 \longrightarrow XCO_2 + H_2O$

7.5 ml 36 ml

$$36 - 7.5 \left(x + \frac{y}{4}\right) + 7.5x = 28.5$$

$$36 - 7.5 \left(15 + \frac{y}{4}\right) + 7.5x = 28.5$$

$$y = 4$$

$$x = 2$$

So, formula = C_2H_4

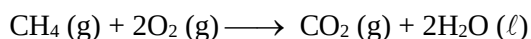
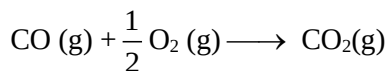
Ex.60 A 30 c.c. mixture of CO, CH_4 and He gases is exploded by an electric discharge at room temperature with excess of oxygen. The decrease in volume is found to be 13 mL. A further contraction of 14 mL occurs, when the residual gas is treated with KOH solution. Find out the composition of the gaseous mixture in terms of volume percentage.

Ans. Percentage composition of CO = 33.33 %; CH_4 = 13.33 %; He = 53.33 %

Sol. Let the volume of CO be 'a' mL and CH_4 be 'b' mL

$$\therefore \text{Volume of He} = (30 - a - b)$$

on explosion with oxygen



'a' mL of CO give 'a' mL of CO_2 and 'b' mL of CH_4 gives 'b' mL of CO_2 .

Therefore, the volume decrease is due to the consumption of O_2 . O_2 consumed for 'a' mL of CO is $\frac{a}{2}$ mL

and O_2 consumed for 'b' mL of CH_4 is '2b' mL

$$\therefore \frac{a}{2} + 2b = 13$$

The further contraction occurs because of the absorption of CO_2 by KOH, $a + b = 14$

$$\therefore b = 4 \text{ mL}$$

$$\therefore a = 10 \text{ mL}$$

$$\therefore \text{Percentage composition of CO} = \frac{10}{30} \times 100 = 33.33 \%$$

$$\text{Percentage composition of } CH_4 = \frac{4}{30} \times 100 = 13.33$$

$$\text{Percentage composition of He} = \frac{(30 - 10 - 4)}{30} \times 100 = 53.33 \%$$

EXERCISE #1

Dalton's Atomic Theory

- Which postulate of Dalton's Atomic theory suggests the law of conservation of mass?
 - Atoms cannot be divided.
 - All the matter is made up of very small particles called atoms.
 - Elements consists of atoms combined in a fixed ratio.
 - Atoms can neither be created nor be destroyed.
- Dalton incorporated the law of conservation of mass into his atomic theory by asserting that-
 - Matter is composed of atom
 - Atom can be destroyed in the chemical reactions.
 - Atoms are indivisible.
 - Atoms can be combined in any ratio of their mass.
- Dalton's Atomic theory agrees with modern atomic theory except for the statement that-
 - Atoms are not divided in chemical reactions.
 - all atoms of same elements have same mass.
 - Atoms of the same element are chemically alike.
 - Atoms can be combined in any ratio of their mass.

Laws of Chemical Combination

- All samples of carbon dioxide contain carbon and oxygen in the mass ratio 3: 8. This is in agreement with the law of

(A) conservation of mass	(B) constant proportions
(C) (A) and (B) both	(D) none of these
- A chemical equation is always balanced to fulfil the condition of -

(A) Law of conservation of mass.	(B) Law of multiple proportions.
(C) Law of constant proportions	(D) All of these
- The law of conservation of mass was given by

(A) John Dalton	(B) Proust	(C) Lavoisier	(D) None of these
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- If the mass of reactants is equal to the mass of products then which of the following statements is true?
 - Law of conservation of mass holds good.
 - Mass can neither be created nor be destroyed.
 - There is no change in mass during a chemical reaction.
 - All of the above
- Hydrogen & Oxygen combine in the ratio of 1: 8 by mass to form water. What mass of oxygen gas would be required to react completely with 3 g of hydrogen gas?

(A) 24 g	(B) 27 g	(C) 21 g	(D) 3 g
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- Ammonia contains 82.65% N_2 and 17.65% H_2 . If law of constant proportion is true, then mass of zinc required to give 10g Ammonia will be:

(A) 8.265 g	(B) 0.826 g	(C) 82.65 g	(D) 82.65 g
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7. Irrespective of source, pure H_2O sample of H_2O always yields 88.89% mass of oxygen and 11.11% mass of hydrogen. This is explained by law of -
 (A) conservation of mass (B) constant composition
 (C) multiple proportion (D) constant volume
8. One part of an element A combines with 2 parts of another element B. 6 parts of element C combine with 4 parts of element B. If A & C combine together, the ratio of their masses will be governed by:
 (A) law of definite proportions (B) law of multiple proportions
 (C) law of reciprocal proportions (D) law of conservation of mass.
9. How much volume of oxygen will be required for complete combustions of 40ml of acetylene and how much volume of carbon dioxide will be formed. All volume measured at NTP.
 (A) 100, 50 (B) 100, 60 (C) 100, 80 (D) 100, 40
10. What would be the mass of water formed by reaction of 2 grams hydrogen with 16g oxygen?
 (A) 16 g (B) 32 g (C) 18 g (D) 20 g
11. Hydrogen combines with chlorine to form HCl . It also combines with sodium to form NaH . If sodium and chlorine also combine with each other, they will do so in ratio of masses.
 (A) 23:35.5 (B) 35.5:23 (C) 1:1 (D) 23:1
12. The law of multiple proportions is illustrated by pair of compounds:
 (A) NaCl , NaBr (B) H_2O , D_2O (C) SO_2 , SO_3 (D) HgOH , HgO
13. A sample of CaCO_3 has following % composition: Ca = 40%, C = 12%, O = 48%. If law of constant proportions is true, then weight of Ca in 4g of a sample of calcium carbonate obtained from another source will be.
 (A) 0.016 g (B) 0.16 g (C) 1.6 g (D) 16 g
14. 4.8 g of KClO_3 when heated produced 1.9 g of oxygen and the residue (KCl) left behind 2.9 g. Which law of chemical combination is illustrated?
 (A) law of constant proportion (B) law of multiple proportion
 (C) law of conservation of mass. (D) law of reciprocal proportion
15. H_2O_2 and H_2O contains 5.93% and 11.2% of H_2 respectively. Which law of chemical combination is:
 (A) law of constant proportion (B) law of multiple proportion
 (C) law of conservation of mass. (D) law of reciprocal proportion

Percentage Composition

1. The volume percentage of the solute in 250 ml of a solution containing 5gm of sugar is
 (A) 4% (B) 2% (C) 5% (D) 9%
2. The amount of glucose required to prepare a 250gm of 5% solution of glucose by mass is
 (A) 12.5g (B) 15g (C) 17g (D) 20g
3. Calculate the percentage composition of calcium in calcium nitrate.
 (A) 24% (B) 12% (C) 55% (D) 9%

4. Calculate the percentage composition of water of crystallisation in $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$
 (A) 45.3% (B) 52% (C) 25% (D) 19%
5. It is found that 16.5 g of metal combine with oxygen to form 35.60 g of metal oxide. Calculate the percentage of metal and oxygen in the compound.
 (A) 46, 54% (B) 20, 80% (C) 55, 45% (D) 25, 75%
6. Density of dry air containing only N_2 and O_2 is 1.15g/L at 740 mm and 300K. What is % composition of N_2 by weight in the air?
 (A) 78% (B) 75.5% (C) 70.02% (D) 73.42%
7. A sample of calcium carbonate (CaCO_3) has the following percentage composition: Ca = 40%, C = 12%, O = 48%. If the law of constant proportions is true, then the weight of calcium in 4g of a sample of calcium carbonate obtained from another source will be:
 (A) 0.016 g (B) 0.16 g (C) 1.6 g (D) 16 g
8. What percentage of oxygen is present in the compound $\text{CaCO}_3 \cdot 3\text{Ca}_3(\text{PO}_4)_2$?
 (A) 23.3% (B) 45.36% (C) 41.94% (D) 17.08%
9. 2.0 g sample contain mixture of SiO_2 and Fe_2O_3 , on very strong heating leave a residue weighing 1.96 g. The reaction responsible for loss of weight is

$$\text{Fe}_2\text{O}_3 (\text{s}) \longrightarrow \text{Fe}_3\text{O}_4 (\text{s}) + \text{O}_2 (\text{g}), \text{ (unbalance equation)}$$

 What is the percentage by mass of SiO_2 in original sample?
 (A) 10% (B) 20% (C) 40% (D) 60%
10. The weight of a mixture containing HCl and H_2SO_4 is 0.1g. On treatment with an excess of an AgNO_3 solution, reacted with this acid mixture gives 0.1435g of AgCl. Weight % of the H_2SO_4 is mixture is:
 (A) 36.5 (B) 63.5 (C) 50 (D) None of these

Mole Concept

1. Which of the following has the smallest mass?
 (A) 4 g of He (B) 6.023×10^{23} atoms of He
 (C) 1 atom of He (D) 1 mole atoms of He
2. The number of carbon atoms in 1 g of CaCO_3 is:
 (A) 6.023×10^{23} (B) 6.023×10^{21} (C) 3.0125×10^{22} (D) 1.204×10^{23}
3. 6.023×10^{20} atoms of silver (Atomic mass = 108 u) weigh:
 (A) 108×10^3 g (B) 108 g (C) 0.108 g (D) 10.8 g
4. Which of the following has largest number of molecules?
 (A) 8 g of CH_4 (B) 4.4 g of CO_2
 (C) 34.2 g of $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ (D) 2 g of H_2
5. Which of the following contains one mole molecules of the substance?
 (A) 16 g Oxygen (B) 7 g Nitrogen (C) 2 g Hydrogen (D) 36 g Water.

6. The number of molecules in 16.0 g of oxygen is:
(A) 6.02×10^{23} (B) 6.02×10^{-23} (C) 3.01×10^{-23} (D) 3.01×10^{23}
7. The volume of one mole of a gas at normal temperature and pressure is:
(A) 11.2 litres (B) 22.4 litres (C) 100 litres (D) None of these
8. The number of gram atoms in 8 g of He are:
(A) 2 (B) 1.204×10^{24} (C) 3.1×10^{23} (D) None of these
9. The percentage of hydrogen in H_2O is:
(A) 8.88 (B) 11.12 (C) 20.60 (D) 80.0
10. The charge in coulombs of 1 gram ion of N^{3-} is (the charge on an electron is 1.602×10^{-19} C):
(A) 2.894×10^5 C (B) 3.894×10^5 C (C) 2.894×10^6 C (D) None of these
11. The mass of oxygen contained in 1 kg of potassium nitrate (KNO_3) will be:
(A) 478.5 g (B) 485.5 g (C) 475.2 g (D) 488.2 g
12. 1 mole of a compound contains 1 mole of carbon and 2 moles of oxygen. The molecular weight of the compound is:
(A) 3 (B) 12 (C) 32 (D) 44
13. What amount of CaO will be produced by 1 g of calcium?
(A) 2.4 g (B) 1.4 g (C) 2.1 g (D) 3 g
14. A sample of KClO_3 on decomposition yielded 448 mL of oxygen gas at NTP. Calculate the moles of oxygen produced:
(A) 0.02 (B) 0.03 (C) 0.2 (D) 2
15. In Haber process 30 litres of dihydrogen and 30 litres of dinitrogen were taken for reaction which yielded only 50% of the expected product. What will be the volume of dinitrogen and dihydrogen consumed during the reaction?
(A) $\text{H}_2 = 10\text{L}$, $\text{N}_2 = 5\text{L}$ (B) $\text{N}_2 = 30\text{L}$, $\text{H}_2 = 5\text{L}$
(C) $\text{N}_2 = 15\text{L}$, $\text{H}_2 = 5\text{L}$ (D) $\text{N}_2 = 10\text{L}$, $\text{H}_2 = 50\text{L}$
16. The weight of oxygen required for the complete combustion of 2.8 g of ethylene is
(A) 9.6 g (B) 1.4 g (C) 2.1 g (D) 3 g
17. Which of the following sample contains the maximum number of atoms -
(A) 1 mg of C_4H_{10} (B) 1 mg of N_2 (C) 1 mg of Na (D) 1 mL of water
18. The total number of protons, electrons and neutrons in 12 g of ${}^{12}_6\text{C}$ is -
(A) 1.084×10^{25} (B) 6.022×10^{23} (C) 6.022×10^{22} (D) 18
19. 4.4 g of CO_2 and 2.24 litre of H_2 at STP are mixed in a container. The total number of molecules present in the container will be -
(A) 6.022×10^{23} (B) 1.2044×10^{23} (C) 2 mole (D) 6.023×10^{24}

20. How many atoms are contained in a mole of Ca(OH)_2 :
 (A) $30 \times 6.02 \times 10^{23}$ atoms/mol (B) $5 \times 6.02 \times 10^{23}$ atoms/mol
 (C) $6 \times 6.02 \times 10^{23}$ atoms/mol (D) None of these
21. Which of the following has the maximum mass?
 (A) 25 g of Hg (B) 2 moles of H_2O (C) 2 moles of CO_2 (D) 4 g atom of oxygen
22. If the atomic mass of Sodium is 23, the number of moles in 46 g of sodium is:
 (A) 1 (B) 2 (C) 2.3 (D) 4.6
23. Number of moles present in 1 m^3 of a gas at NTP are:
 (A) 44.6 (B) 40.6 (C) 42.6 (D) 48.6
24. The volume occupied by one molecule of water (density 1 g/cm^3) is:
 (A) 18 cm^3 (B) 22400 cm^3 (C) 6.023×10^{-23} (D) $3.0 \times 10^{-23} \text{ cm}^3$
25. Total mass of neutrons in 7mg of ^{14}C is:
 (A) $3 \times 10^{20} \text{ kg}$ (B) $4 \times 10^{-6} \text{ kg}$ (C) $5 \times 10^{-7} \text{ kg}$ (D) $4 \times 10^{-7} \text{ kg}$

Limiting Reagent

1. If 0.5 mol of BaCl_2 is mixed with 0.1 mole of Na_3PO_4 , the maximum number of mole of $\text{Ba}_3(\text{PO}_4)_2$ that can be formed is:
 (A) 0.7 (B) 0.05 (C) 0.30 (D) 0.10
2. For the reaction $2\text{P} + \text{Q} \rightarrow \text{R}$, 8 mol of P and 5 mol of Q will produce:
 (A) 8 mol of R (B) 5 mol of R (C) 4 mol of R (D) 13 mol of R
3. 4 g of hydrogen is ignited with 4 g of oxygen, the amount of water formed is?
 (A) 2.5 g (B) 0.5 g (C) 4.5 g (D) 8 g
4. 0.5 mole of H_2SO_4 is mixed with 0.2 mole of Ca(OH)_2 . The maximum number of moles of CaSO_4 formed is:
 (A) 0.2 (B) 0.5 (C) 0.4 (D) 1.5
5. For the reaction: $\text{A} + 2\text{B} \rightarrow \text{C}$
 5 mole of A and 8 mole of B will produce:
 (A) 5 mole of C (B) 4 mole of C (C) 8 mole of C (D) 12 mole of C
6. Zinc and hydrochloric acid react according to the reaction:
 $\text{Zn(s)} + 2\text{HCl(aq.)} \longrightarrow \text{ZnCl}_2\text{(aq.)} + \text{H}_2\text{(g)}$
 If 0.3 mole of Zn are added to hydrochloric acid containing 0.52 mole HCl, how many moles of H_2 are produced:
 (A) 0.26 (B) 0.52 (C) 0.14 (D) 0.30
7. In Haber process, 30 litres of dihydrogen and 30 litres of dinitrogen were taken for reaction which yielded only 50% of the expected product. What will be the composition of the gaseous mixture under the aforesaid condition in the end?
 (A) 20 litres NH_3 , 25 litres N_2 , 20 litres H_2 (B) 10 litres NH_3 , 25 litres N_2 , 15 litres H_2
 (C) 20 litres NH_3 , 10 litres N_2 , 30 litres H_2 (D) 20 litres NH_3 , 25 litres N_2 , 15 litres H_2

Concentration Terms

- 100 mL of 1 M HCl is mixed with 50 mL of 2 M HCl. Hence, $[H_3O^+]$ is:
(A) 1.00 M (B) 1.50 M (C) 1.33 M (D) 3.00 M
- The mass percentage of sugar in a solution containing 47g of sugar and 72g of water is:
(A) 40% (B) 30% (C) 11% (D) 7%
- The mass percentage of sugar in a solution containing 100g of sugar and 100g of water is:
(A) 40% (B) 50% (C) 11% (D) 7%
- 4.9 g of H_2SO_4 is present in 100 mL of the solution. What is the molarity of the solution?
(A) 0.2M (B) 0.5M (C) 0.21M (D) 0.25M
- What volume of a 0.8 M solution contains 100 millimoles of the solute?
(A) 80 mL (B) 125 mL (C) 125 L (D) 80 L
- Calculate the molarity when:
(a) 4.9 g H_2SO_4 acid dissolved in water to result 500 mL solution.
(b) 2 gram-molecules of KOH dissolved in water to result 500 mL solution.
(A) (a) 0.1 M (b) 0.07 M (B) (a) 0.4 M (b) 4 M
(C) (a) 0.4 M (b) 0.07 M (D) (a) 0.1 M (b) 4 M
- Calculate the amount of 75% pure NaI required to prepare 5 litre of 0.5 M solution.
(A) 281.25 g (B) 500 g (C) 923.33 g (D) 519.375 g
- Concentrated HNO_3 is 63% HNO_3 by mass and has a density of 1.4 g/mL. How many millilitres of this solution are required to prepare 250 mL of a 1.20 M HNO_3 solution?
(A) 18.0 (B) 21.42 (C) 20.0 (D) 14.21
- What value of 75% alcohol by weight ($d = 0.80$ g/cm³) must be used to prepare 150 cm³ of 30% alcohol by weight ($d = 0.90$ g/cm³)?
(A) 67.5 mL (B) 56.25 mL (C) 44.44 mL (D) None of these
- H_3PO_4 (98 g mol⁻¹) is 98% by mass of solution. If the density is 1.8 g/mL, the molarity is
(A) 18 M (B) 36 M (C) 54 M (D) 0.18 M
- How would you prepare exactly 3 litre of 1 M NaOH solution by mixing proportions of stock solutions of 2.5 M NaOH and 0.4 M NaOH, if no water is to be used. Find the ratio of the volume (V_1/V_2).
(A) 1: 3 (B) 3: 7 (C) 2: 5 (D) Data insufficient
- What approximate volume of 0.40 M $Ba(OH)_2$ solution must be added to 50.0 mL of 0.30 M the molarity of the OH^- ions is 0.50 M ?
(A) 33 mL (B) 66 mL (C) 133 mL (D) 100 mL
- 49 gram of H_2SO_4 is present in 67 g of solution mole fraction of water will be:
(A) 0.33 (B) 0.66 (C) 0.6 (D) 0.5

14. 50 g of 40% w/w NaOH contain. Calculate mole fraction of water?
(A) 0.23 (B) 0.76 (C) 0.3 (D) 0.7
15. 100 ml of 80% w/v NaOH contain. Calculate mole fraction of NaOH?
(A) 0.64 (B) 0.36 (C) 0.23 (D) 0.76
16. 100 g of water is added to 0.585 g of NaCl what is the molarity of $\text{Cl}^\ominus$ ion?
(A) 10^{-4} M (B) 10^{-1} M (C) 10^{-2} M (D) 10^{-3} M
17. In 2 mole of solution mole fraction of water is 0.8. Moles of solute will be?
(A) 0.4 (B) 1.6 (C) 0.8 (D) 1.2
18. 40.25 g of Glauber's salt is dissolved in water to obtain 500 mL of solution of density 1077.5 g dm^{-3} . The molality of Na_2SO_4 in solution is about
(A) 0.35 mol kg^{-1} (B) 0.24 mol kg^{-1} (C) 0.46 mol kg^{-1} (D) 0.17 mol kg^{-1}
19. The mole fraction of ethanol in water is 0.08. Its molality will be
(A) 2.41 mol kg^{-1} (B) 4.83 mol kg^{-1} (C) 3.33 mol kg^{-1} (D) 6.41 mol kg^{-1}

EXERCISE #2

- A 20.00 mL sample of a $\text{Ba}(\text{OH})_2$ solution is titrated with 0.245 M HCl. If 27.15 mL of HCl is required, what is the molarity of the $\text{Ba}(\text{OH})_2$ solution?
(A) 0.166 M (B) 0.180 M (C) 0.333 M (D) 0.666 M
- How many moles of Na^+ ions are in 20 mL of 0.40 M Na_3PO_4 ?

Molar Mass, M
$\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$ 666.43 g·mol

 (A) 0.0080 (B) 0.024 (C) 0.050 (D) 0.20
- What is the mass percent of oxygen in $\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$?
(A) 9.60 (B) 28.8 (C) 43.2 (D) 72.0
- What is the number of O_2 molecules in the 2.5 g of O_2 inhaled by the average person in one minute?
(A) 1.9×10^{22} (B) 3.8×10^{22} (C) 4.7×10^{22} (D) 9.4×10^{22}
- If 1.50 g of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ were heated to drive off the water of hydration, how much anhydrous $\text{H}_2\text{C}_2\text{O}_4$ would remain?
(A) 0.34 g (B) 0.92 g (C) 1.07 g (D) 1.50 g
- How many mL of 8.00 M HCl are needed to prepare 150. mL of a 1.60 M HCl solution?
(A) 30.0 mL (B) 24.0 mL (C) 18.8 mL (D) 12.0 mL
- Analysis of a compound known to contain only Mg, P, and O gives this analysis. 21.8% Mg 27.7% P 50.3% O. What is its empirical formula?
(A) MgPO_2 (B) MgPO_3 (C) $\text{Mg}_2\text{P}_2\text{O}_7$ (D) $\text{Mg}_3\text{P}_2\text{O}_8$
- The reaction of ethanol, $\text{C}_2\text{H}_5\text{OH}$, with oxygen is a popular classroom demonstration. Balance the equation to find the number of moles of gaseous products formed per mole of ethanol.

$$\text{C}_2\text{H}_5\text{OH}(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$$
 (A) 2 (B) 3 (C) 4 (D) 5
- Ammonia is produced in accordance with this equation.

$$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$$
 In a particular experiment, 0.25 mol of NH_3 is formed when 0.5 mol of N_2 is reacted with 0.5 mol of H_2 . What is the percent yield?
 (A) 75% (B) 50% (C) 33% (D) 25%
- A chloride salt of rhenium contains 63.6% Re by mass. What is its empirical formula?
(A) ReCl (B) ReCl_2 (C) ReCl_3 (D) ReCl_5
- The amount of chloride ion in a water sample is to be determined by adding excess silver nitrate. If 1.0 g of silver chloride is precipitated, what mass of chloride ion is in the original sample?
(A) 0.25 g (B) 0.34 g (C) 0.50 g (D) 0.75 g

12. How many millimoles of methane, CH_4 , are present in 6.4 g of this gas?
 (A) 0.40 (B) 4.0 (C) 40 (D) 4.0×10^2
13. According to the equation $\text{SnO}_2 + 2\text{H}_2 \rightarrow \text{Sn} + 2\text{H}_2\text{O}$.
 What volume of hydrogen, measured at 1 atm and 273 K, is required to react with 2.00 g of SnO_2 ?
 (A) 0.00133 L (B) 0.00265 L (C) 0.297 L (D) 0.595 L
14. An oxide of manganese contains 2.29 g of manganese per gram of oxygen. What is the empirical formula of this compound?
 (A) MnO (B) MnO_2 (C) Mn_2O_3 (D) MnO_3
15. A heterogeneous system is produced when 0.040 moles of solid NaCl is added to 0.10 L of 0.10 M $\text{Pb}(\text{NO}_3)_2$. Which ion is present in the aqueous phase at the highest concentration?
 (A) Cl^- (B) NO_3^- (C) Pb^{2+} (D) Na^+
16. Which expression gives the fraction by mass of nitrogen in ammonium dihydrogen phosphate?
 (A) 14 / 115 (B) 28 / 115 (C) 28 / 132 (D) 14 / 210
17. Ethanol burns in excess oxygen to form $\text{CO}_2(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ according to this balanced equation.

$$\text{C}_2\text{H}_5\text{OH}(\text{g}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{g})$$

 What value is closest to the volume of $\text{CO}_2(\text{g})$, measured at 200K and 1 atm, produced from the combustion of 0.25 mol of $\text{C}_2\text{H}_5\text{OH}(\text{g})$?
 (A) 5 L (B) 8 L (C) 10 L (D) 15 L
18. Adipic acid, $\text{HOOC}(\text{CH}_2)_4\text{COOH}$, is used in making nylon. What is the total number of atoms in 1.0 g of adipic acid?

Molar Mass, g.mol^{-1}

adipic acid 146.26

- (A) 20 (B) 4.1×10^{21} (C) 8.2×10^{22} (D) 7.2×10^{24}
19. Calculate the mass percentage of nitrogen in hydrazinium sulfate $(\text{N}_2\text{H}_5)_2\text{SO}_4$.

Molar Mass, g.mol^{-1}

$(\text{N}_2\text{H}_5)_2\text{SO}_4$ 162.2

- (A) 10.8 (B) 17.3 (C) 34.5 (D) 51.2
20. Acetylene, C_2H_2 , reacts with oxygen according to the unbalanced equation:

$$\text{C}_2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$$

 What is the $\text{O}_2/\text{C}_2\text{H}_2$ ratio when this equation is correctly balanced?
 (A) 2/1 (B) 3/1 (C) 4/1 (D) 5/2
21. Silicon carbide, SiC , is produced by heating SiO_2 and C to high temperatures according to the equation:

$$\text{SiO}_2(\text{s}) + 3\text{C}(\text{s}) \rightarrow \text{SiC}(\text{s}) + 2\text{CO}(\text{g})$$

 How many grams of SiC could be formed by reacting 2.00 g of SiO_2 and 2.00 g of C ?
 (A) 1.33 (B) 2.26 (C) 3.59 (D) 4.00

22. A 7.66 g sample of hydrated sodium sulfate, $\text{Na}_2\text{SO}_4 \cdot x\text{H}_2\text{O}$, forms 4.06 g of anhydrous Na_2SO_4 . What is the value of x ?

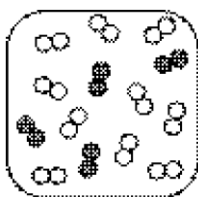
Molar Mass, $\text{g} \cdot \text{mol}^{-1}$	
Na_2SO_4	142

- (A) 0.2 (B) 3.6 (C) 5 (D) 7
23. Silver metal reacts with nitric acid according to the equation:

$$3\text{Ag(s)} + 4\text{HNO}_3\text{(aq)} \rightarrow 3\text{AgNO}_3\text{(aq)} + \text{NO(g)} + 2\text{H}_2\text{O(l)}$$
 What volume of 1.15 M $\text{HNO}_3\text{(aq)}$ is required to react with 0.784 g of silver?
 (A) 4.74 mL (B) 6.32 mL (C) 8.43 mL (D) 25.3 mL
24. How many water molecules are in a 0.10 g sample of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (MM = 249.7)?
 (A) 1.2×10^{21} (B) 2.4×10^{21} (C) 2.4×10^{22} (D) 1.2×10^{23}
25. Acetylene, C_2H_2 , reacts with O_2 to produce CO_2 and H_2O . What is the $\text{O}_2/\text{C}_2\text{H}_2$ ratio in the balanced equation?
 (A) 2/1 (B) 3/2 (C) 5/2 (D) 3/1
26. How many moles of stomach acid can be neutralized by Mg(OH)_2 in the form of Milk of Magnesia is used to neutralize excess stomach acid 1.00 g of Mg(OH)_2 ?
 (A) 0.0171 (B) 0.0343 (C) 0.686 (D) 1.25
27. A 25.00 mL sample of 0.1050 M H_2SO_4 is titrated with a NaOH solution of unknown concentration. The phenolphthalein endpoint was reached when 17.23 mL of the NaOH solution had been added. What is the concentration of the NaOH?
 (A) 0.07617 M (B) 0.1447 M (C) 0.1524 M (D) 0.3047 M
28. A sample of oxygen gas and a sample of an unknown gas are weighed separately in the same evacuated flask. Use the data given to find the molar mass of the unknown gas (assume experiments are carried out at the same pressure and temperature).

Mass of evacuated flask	124.46 g
Mass of flask + oxygen	125.10 g
Mass of flask + unknown gas	125.34 g

- (A) 22 g/mol (B) 38 g/mol (C) 44 g/mol (D) 84 g/mol
29. What is the mass of one molecule of water in grams?
 (A) 18 (B) 1.1×10^{-21} (C) 3.0×10^{-23} (D) 1.7×10^{-24}
30. If nitrogen atoms are represented as filled circles and oxygen atoms as open circles, how much NO_2 can be prepared from the mixture shown?



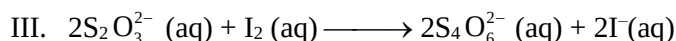
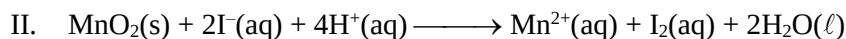
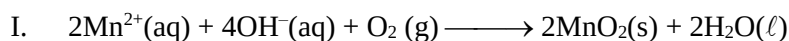
- (A) 4 molecules (B) 5 molecules (C) 6 molecules (D) 8 molecules

31. Naturally occurring thallium consists of two stable isotopes, Tl-203 and Tl-205 (atomic masses = 203.0 and 205.0, respectively) and has an average atomic mass of 204.4. What is the percentage of Tl-205?
(A) 14.0% (B) 30.0% (C) 50.0 % (D) 70.0 %
32. What is the maximum mass (in grams) of NO that could be obtained from 15.5 g of N_2O_4 and 4.68 g of N_2H_4 when they react? The balanced chemical equation is;
$$2\text{N}_2\text{O}_4 + \text{N}_2\text{H}_4 \rightarrow 6\text{NO} + 2\text{H}_2\text{O}.$$

(A) 4.38 (B) 5.04 (C) 15.2 (D) 26.2
33. What volume of 0.108 M H_2SO_4 is required to neutralize 25.0 mL of 0.145 M KOH?
(A) 16.8 mL (B) 33.6 mL (C) 37.2 mL (D) 67.1 mL
34. A mineral containing iron(II) sulfide but no other sulfides is treated with excess hydrochloric acid to produce hydrogen sulfide. If a 3.15 g sample of the mineral yields 448 mL of hydrogen sulfide gas (measured at 0 °C and 760 mm Hg), what is the mass percentage of iron(II) sulfide in the sample?
(A) 20.4 (B) 35.5 (C) 55.8 (D) 71.0
35. Iodine adds to the double bonds in fatty acids (one iodine molecule per double bond). How many double bonds are in a molecule of arachidonic acid (Molar mass = 304.5 g/mol) if 0.125 g of the acid require 0.417 g of iodine?
(A) 2 (B) 3 (C) 4 (D) 8

Question number 36, 37, 38 and 39 are based on the following information:

Dissolved oxygen in water is determined by using a redox reaction. Following equations describe the procedure -



36. How many moles of $2\text{S}_2\text{O}_3^{2-}$ are equivalent to each mole of O_2 ?
(A) 0.5 (B) 1 (C) 2 (D) 4
37. What amount of I_2 will be liberated from 8 g dissolved oxygen?
(A) 127 g (B) 254 g (C) 504 g (D) 1008 g
38. 3×10^{-3} moles O_2 is dissolved per litre of water, then what will be molarity of I^{-} produced in the given reaction?
(A) 3×10^{-3} M (B) $4 \times 3 \times 10^{-3}$ M (C) $2 \times 3 \times 10^{-3}$ M (D) $\frac{1}{2} \times 3 \times 10^{-3}$ M
39. 8 mg dissolved oxygen will consume:
(A) 5×10^{-4} mol Mn^{+2} (B) 2.5×10^{-4} mol Mn^{2+}
(C) 10 mol Mn^{2+} (D) 2 mol Mn^{2+}

40. The equation: $2\text{Al}(\text{S}) + \frac{3}{2} \text{O}_2(\text{g}) \longrightarrow \text{Al}_2\text{O}_3(\text{S})$ show that:
- (A) 2 mole of Al reacts with $\frac{3}{2}$ mole of O_2 to produce $\frac{7}{2}$ mole of Al_2O_3
- (B) 2g of Al reacts with $\frac{3}{2}$ g of O_2 to produce one mole of Al_2O_3
- (C) 2g mole of Al reacts with $\frac{3}{2}$ litre of O_2 to produce 1 mole of Al_2O_3
- (D) 2 mole of Al reacts with $\frac{3}{2}$ mole of O_2 to produce 1 mole of Al_2O_3
41. Volume of CO_2 obtained at STP by the complete decomposition of 9.85 g. BaCO_3 is:
(At. wt. of Ba = 137)
- (A) 2.24 lit (B) 1.12 lit (C) 0.84 lit (D) 0.56 lit
42. The volume of a gas in discharge tube is 1.12×10^{-7} ml. at STP then the number of molecule of gas in the tube is:
- (A) 3.01×10^4 (B) 3.01×10^{15} (C) 3.01×10^{12} (D) 3.01×10^{16}
43. A person adds 1.71 gram of sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in order to sweeten his tea. The number of carbon atoms added are (mol. mass of sugar = 342)
- (A) 3.6×10^{22} (B) 7.2×10^{21} (C) 0.05 (D) 6.6×10^{22}
44. The volume of CO_2 (in litres) liberated at STP when 10 g of 90% pure limestone is heated completely, is:
- (A) 22.4 L (B) 2.24 L (C) 20.16 L (D) 2.016 L
45. The number of atoms of Cr and O in a compound are 4.8×10^{10} & 9.6×10^{10} respectively. Its empirical formula is:
- (A) Cr_2O_3 (B) CrO_2 (C) Cr_2O_4 (D) None
46. A compound of X and Y has equal mass of them. If their atomic weights are 30 and 20 respectively. Molecular formula of that compound (its mol wt. is 120) could be:
- (A) X_2Y_2 (B) X_3Y_3 (C) X_2Y_3 (D) X_3Y_2
47. 1 litre of a hydrocarbon weights as much as one litre of CO_2 . Then the molecular formula of the hydrocarbon is:
- (A) C_3H_8 (B) C_2H_6 (C) C_2H_4 (D) C_3H_6
48. In the reaction, $\text{CrO}_5 + \text{H}_2\text{SO}_4 \longrightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O} + \text{O}_2$. One mole of CrO_5 will liberate how many moles of O_2 ?
- (A) 5/2 (B) 5/4 (C) 9/2 (D) None
49. The moles of O_2 required for reacting with 6.8 g of ammonia ($\text{.....NH}_3 + \text{.....O}_2 \longrightarrow \text{.....NO} + \text{....H}_2\text{O}$) is:
- (A) 5 (B) 2.5 (C) 1 (D) 0.5

50. The hydrated salt, $\text{Na}_2\text{SO}_4 \cdot n\text{H}_2\text{O}$ undergoes 55.9% loss in weight on heating and becomes anhydrous. The value of n will be:
 (A) 5 (B) 3 (C) 7 (D) 10
51. How many gram of KCl would have to be dissolved in 60 g of H_2O to give 20% by weight of solution -
 (A) 15 g (B) 1.5 g (C) 11.5 g (D) 31.5 g
52. If V ml of the vapours of substance at NTP weight W g. Then mol w. of substance is:
 (A) $(W/V) \times 22400$ (B) $\frac{V}{W} \times 22.4$ (C) $(W-V) \times 22400$ (D) $\frac{W \times 1}{V \times 22400}$
53. The molality of a sulphuric acid solution is 0.2. Calculate the total weight of the solution having 1000 gm of solvent.
 (A) 1000 g (B) 1098.6 g (C) 980.4 g (D) 1019.6g
54. The total number of AlF_3 molecule in a sample of AlF_3 containing 3.01×10^{23} ions of F^- is -
 (A) 9.0×10^{24} (B) 3.0×10^{24} (C) 7.5×10^{23} (D) 10^{23}
55. Equal masses of O_2 , H_2 and CH_4 are taken in a container. The respective mole ratio of these gases in container is
 (A) 1: 16: 2 (B) 16: 1: 2 (C) 1: 2: 16 (D) 16: 2: 1
56. The total number of molecules of SO_2Cl_2 in 13.5 g of sulphuryl chloride is
 (A) 0.1 (B) 0.2 (C) 0.3 (D) 0.4
57. The number of sodium atoms in 2 moles of sodium ferrocyanide ($\text{Na}_4[\text{Fe}(\text{CN})_6]$) is:
 (A) 12×23 (B) 26×10^{23} (C) 34×10^{23} (D) 48×10^{23}
58. The empirical formula of a compound of molecular mass 120 is CH_2O . The molecular formula of the compound is
 (A) $\text{C}_2\text{H}_4\text{O}_2$ (B) $\text{C}_4\text{H}_8\text{O}_1$ (C) $\text{C}_3\text{H}_6\text{O}_3$ (D) all of these
59. A hydrocarbon contains 75% of carbon. Then its empirical formula is:
 (A) CH_4 (B) C_2H_4 (C) C_2H_6 (D) C_2H_2
60. Which of the following compounds has same empirical formula as that of glucose:
 (A) CH_3CHO (B) CH_3COOH (C) CH_3OH (D) C_2H_6
61. Caffeine has a molecular weight of 194. It contains 28.9% by mass of nitrogen number of atoms of nitrogen in one molecule of it:
 (A) 2 (B) 3 (C) 4 (D) 5
62. 10g of piece of marble was put into excess of dilute HCl acid. When reaction was complete 1120 cm^3 of CO_2 was obtained at STP. The evaluate the value of $x/10$ where $x = \%$ of CaCO_3 in marble.
 (A) 4 (B) 5 (C) 8 (D) 10

63. Vapour density of a gas if its density is 0.178 g/L at NTP is:
 (A) 0.178 (B) 2 (C) 4 (D) 0.089
64. Vapour density of a volatile substance w.r.t. CH_4 is 4 ($\text{CH}_4 = 1$). Its molecular weight would be:
 (A) 8 (B) 32 (C) 64 (D) 128
65. A gas is found to have the formula $(\text{CO})_x$. It's VD is 70 the value of x must be:
 (A) 7 (B) 4 (C) 5 (D) 6
66. 8g of sulphur are burnt to form SO_2 , which is oxidised by Cl_2 water. The solution is treated with BaCl_2 solution. The amount of BaSO_4 precipitated is:
 (A) 1.0 mole (B) 0.5 mole (C) 0.75 mole (D) 0.25 mole
67. Weight of oxygen in Fe_2O_3 and FeO is in the simple ratio of:
 (A) 3: 2 (B) 1: 2 (C) 2: 1 (D) 3: 1
68. Amount of oxygen required for combustion of 1 kg of a mixture of butane and isobutane is:
 (A) 1.8 kg (B) 2.7 kg (C) 4.5 kg (D) 3.58 kg
69. Amount of BaSO_4 formed on mixing the aqueous solution of 2.08 g BaCl_2 and excess of dilute H_2SO_4 is:
 (A) 2.33 g (B) 2.08 g (C) 1.04 g (D) 1.165 g
70. The volume of oxygen required for complete combustion of 20 ml of ethene is:
 (A) 30 ml (B) 60 ml (C) 40 ml (D) 50 ml
71. The volume of gas at NTP produced by 100g of CaC_2 with water:
 (A) 70 litre (B) 35 litre (C) 17.5 litre (D) 22.4 litre
72. Sulphur trioxide is prepared by the following two reactions
 $\text{S}_8(\text{s}) + 8\text{O}_2(\text{g}) \rightarrow 8\text{SO}_2(\text{g})$
 $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$
 How many grams of SO_3 are produced from 1 mol of S_8 ?
 (A) 1280.0 (B) 640.0 (C) 960.0 (D) 320
73. Calculate the amount of H_2 which is left unreacted in the given reaction:
 $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$
 If 8 g of H_2 is mixed with 16 g O_2 ?
 (A) 3 g (B) 6 g (C) 1 g (D) 4 g
74. A mixture of 1.0 mole of Al and 3.0 mole of Cl_2 are allowed to react as:
 $2\text{Al}(\text{s}) + 3\text{Cl}_2(\text{g}) \longrightarrow 2\text{AlCl}_3(\text{s})$
 (a) Which is limiting reagent?
 (b) How many moles of AlCl_3 are formed
 (c) Moles of excess reagent left unreacted is
 (A) (a) Al, (b) 1.0 (c) 1.5 (B) (a) Cl_2 , (b) 2.0 (c) 2.0
 (C) (a) Al, (b) 0.5 (c) 1.5 (D) (a) Cl_2 , (b) 1.0 (d) 1.5

75. Molarity of H_2SO_4 (density 1.8g/mL) is 18M. The molality of this solution is:
 (A) 36 (B) 200 (C) 500 (D) 18
76. The mole fraction of water in 20% (wt. /wt.) aqueous solution of H_2O_2 is:
 (A) $\frac{77}{68}$ (B) $\frac{68}{77}$ (C) $\frac{20}{80}$ (D) $\frac{80}{20}$
77. When FeCl_3 is ignited in an atmosphere of pure oxygen, the following reaction takes place

$$4\text{FeCl}_3(\text{s}) + 3\text{O}_2(\text{g}) \longrightarrow 2\text{Fe}_2\text{O}_3(\text{s}) + 6\text{Cl}_2(\text{g})$$

 If 3 moles of FeCl_3 are ignited in the presence of 2 moles of O_2 gas, how much of which reagent is present in excess and therefore, remains unreacted?
 (A) 0.33 mole FeCl_3 remains unreacted (B) 0.67 mole FeCl_3 remains unreacted
 (C) 0.25 mole O_2 remains unreacted (D) 0.50 mole O_2 remains unreacted
78. The molarity of 98% H_2SO_4 ($d = 1.8\text{g/mL}$) by wt. is:
 (A) 6 M (B) 18.74 M (C) 10 M (D) 4 M
79. If 500 ml of 1 M solution of glucose is mixed with 500 ml of 1 M solution of glucose final molarity of solution will be:
 (A) 1 M (B) 0.5 M (C) 2 M (D) 1.5 M
80. What volume of a 0.8 M solution contains 100 millimoles of the solute?
 (A) 100 mL (B) 125 mL (C) 500 mL (D) 62.5 mL
81. 500 mL of a glucose solution contains 6.02×10^{22} molecules. The concentration of the solution is:
 (A) 0.1 M (B) 1.0 M (C) 0.2 M (D) 2.0 M
82. What is the concentration of nitrate ions if equal volumes of 0.1 M AgNO_3 and 0.1 M NaCl are mixed together:
 (A) 0.1 M (B) 0.2 M (C) 0.05 M (D) 0.25 M
83. The molarity of the solution containing 2.8% mass-volume solution of KOH is:
 (A) M/10 (B) M/2 (C) M/5 (D) 1 M
84. The mole fraction of water in a solution containing 117 g sodium chloride and 900 g of water is?
 (A) 0.0632 (B) 0.038 (C) 0.9615 (D) 1.000
85. 250 ml of 0.5 M KCl is diluted with water to 500 ml of solution, the number of chloride ions in the resulting solution are:
 (A) 6.02×10^{23} (B) 3.76×10^{22} (C) 1×10^{24} (D) 3.76×10^{23}

EXERCISE #3 (JEE MAIN)

- In an organic compound of molar mass 108 g mol^{-1} C, H and N atoms are present in 9 : 1 : 3.5 by weight. Molecular formula can be: [AIEEE 2002, 3/225]
 (A) $\text{C}_6\text{H}_8\text{N}_2$ (B) $\text{C}_7\text{H}_{10}\text{N}$ (C) $\text{C}_5\text{H}_6\text{N}_3$ (D) $\text{C}_4\text{H}_{18}\text{N}_3$
- When KMnO_4 acts as an oxidising agent and ultimately forms MnO_4^{2-} , MnO_2 , Mn_2O_3 and Mn^{2+} , then the number of electrons transferred in each case is: [AIEEE 2002, 3/225]
 (A) 4, 3, 1, 5 (B) 1, 5, 3, 7 (C) 1, 3, 4, 5 (D) 3, 5, 7, 1
- Which of the following is a redox reaction? [AIEEE 2002, 3/225]
 (A) $\text{NaCl} + \text{KNO}_3 \longrightarrow \text{NaNO}_3 + \text{KCl}$
 (B) $\text{CaC}_2\text{O}_4 + 2 \text{HCl} \longrightarrow \text{CaCl}_2 + \text{H}_2\text{C}_2\text{O}_4$
 (C) $\text{Mg}(\text{OH})_2 + 2 \text{NH}_4\text{Cl} \longrightarrow \text{MgCl}_2 + 2 \text{NH}_4\text{OH}$
 (D) $\text{Zn} + 2 \text{AgCN} \longrightarrow 2 \text{Ag} + \text{Zn}(\text{CN})_2$
- Which of the following concentration factor is affected by change in temperature? [AIEEE 2002, 3/225]
 (A) Molarity (B) Molality (C) Mole fraction (D) Weight fraction
- What volume of hydrogen gas at 273 K and 1 atm pressure will be consumed in obtaining 21.6 g of elemental boron (atomic mass = 10.8) from the reduction of boron trichloride by hydrogen- [AIEEE 2003, 3/225]
 (A) 44.8 lit. (B) 22.4 lit. (C) 89.6 lit. (D) 67.2 lit.
- 6.02×10^{20} molecules of urea are present in 100 ml of its solution. The concentration of urea solution is - [AIEEE 2004, 3/225]
 (A) 0.001 M (B) 0.01 M (C) 0.02 M (D) 0.1 M
- The oxidation state of Cr in $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$ is : [AIEEE 2005, 1½/225]
 (A) +3 (B) +2 (C) +1 (D) 0
- Two solution of a substance (non electrolyte) are mixed in the following manner. 480 ml of 1.5M first solution + 520 ml of 1.2M second solution. What is the molarity of the final mixture? [AIEEE 2005, 3/225]
 (A) 2.70 M (B) 1.344 M (C) 1.50 M (D) 1.20 M
- Which of the following chemical reactions depicts the oxidizing behaviour of H_2SO_4 ? [AIEEE-2006, 3/165]
 (A) $2\text{HI} + \text{H}_2\text{SO}_4 \rightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$ (B) $\text{Ca}(\text{OH})_2 + \text{H}_2\text{SO}_4 \rightarrow \text{CaSO}_4 + 2\text{H}_2\text{O}$
 (C) $\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{NaHSO}_4 + \text{HCl}$ (D) $2\text{PCl}_5 + \text{H}_2\text{SO}_4 \rightarrow 2\text{POCl}_3 + 2\text{HCl} + \text{SO}_2\text{Cl}_2$
- How many moles of magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$ will contain 0.25 mole of oxygen atoms? [AIEEE-2006, 3/165]
 (A) 0.02 (B) 3.125×10^{-2} (C) 1.25×10^{-2} (D) 2.5×10^{-2}

11. Density of a 2.05M solution of acetic acid in water is 1.02 g/ml. The molality of the solution is: [AIEEE-2006, 3/165]
 (A) 1.14 mol kg⁻¹ (B) 3.28 mol kg⁻¹ (C) 2.28 mol kg⁻¹ (D) 0.44 mol kg⁻¹
12. In the reaction,
 $2\text{Al}_{(s)} + 6\text{HCl}_{(aq)} \rightarrow 2\text{Al}^{3+}_{(aq)} + 6\text{Cl}^{-}_{(aq)} + 3\text{H}_{2(g)}$ [AIEEE-2007, 3/120]
 (A) 6L HCl_(aq) is consumed for every 3L H₂ produced.
 (B) 33.6 L H_{2(g)} is produced regardless temperature and pressure for every moles that reacts.
 (C) 67.2 L H_{2(g)} at STP is produced for every mole of Al that reacts.
 (D) 11.2 L H_{2(g)} at STP is produced for every mole of HCl_(aq) consumed.
13. The density (in g mL⁻¹) of a 3.60 M sulphuric acid solution that is 29% (H₂SO₄ molar mass = 98 g mol⁻¹) by mass will be: [AIEEE-2007, 3/120]
 (A) 1.22 (B) 1.45 (C) 1.64 (D) 1.88
14. A 5.2 molal aqueous solution of methyl alcohol, CH₃OH, is supplied. What is the mole fraction of methyl alcohol in the solution? [AIEEE-2011, 3/120]
 (A) 0.100 (B) 0.190 (C) 0.086 (D) 0.050
15. The molality of a urea solution in which 0.0100 g of urea, [(NH₂)₂CO] is added to 0.3000 dm³ of water at STP is: [Re. Paper AIEEE-2011, 3/120]
 (A) 5.55 × 10⁻⁴ (B) 33.3 m (C) 3.33 × 10⁻² m (D) 0.555 m
16. The density of a solution prepared by dissolving 120 g of urea (mol. mass = 60 u) in 1000 g of water is 1.15 g/mL. The molarity of this solution is : [AIEEE-2012, 4/120]
 (A) 0.50 M (B) 1.78 M (C) 1.02 M (D) 2.05 M
17. The molarity of a solution obtained by mixing 750 mL of 0.5(M) HCl with 250 mL of 2(M)HCl will be: [JEE(Main)-2013, 4/120]
 (A) 0.875 M (B) 1.00 M (C) 1.75 M (D) 0.975 M
18. Consider the following reaction:
 $x\text{MnO}_4^- + y\text{C}_2\text{O}_4^{2-} + z\text{H}^+ \rightarrow x\text{Mn}^{2+} + 2y\text{CO}_2 + \frac{z}{2}\text{H}_2\text{O}$
 The values of x, y and z in the reaction are, respectively: [JEE(Main)-2013, 4/120]
 (A) 5, 2 and 16 (B) 2, 5 and 8 (C) 2, 5 and 16 (D) 5, 2 and 8
19. In which of the following reactions H₂O₂ acts as a reducing agent? [JEE(Main)-2014, 4/120]
 (a) $\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow 2\text{H}_2\text{O}$ (b) $\text{H}_2\text{O}_2 - 2\text{e}^- \longrightarrow \text{O}_2 + 2\text{H}^+$
 (c) $\text{H}_2\text{O}_2 + 2\text{e}^- \longrightarrow 2\text{OH}^-$ (d) $\text{H}_2\text{O}_2 + 2\text{OH}^- - 2\text{e}^- \longrightarrow \text{O}_2 + 2\text{H}_2\text{O}$
 (A) (a), (b) (B) (c), (d) (C) (a), (c) (D) (b), (d)
20. The molecular formula of a commercial resin used for exchanging ions in water softening is C₈H₇SO₃Na (Mol. wt. 206). What would be the maximum uptake of Ca²⁺ ions by the resin when expressed in mole per gram resin? [JEE(Main)-2015, 4/120]
 (A) $\frac{1}{103}$ (B) $\frac{1}{206}$ (C) $\frac{2}{309}$ (D) $\frac{1}{412}$

[JEE Main-2021-22]

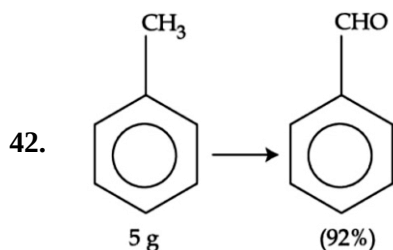
21. 120 of an organic compound that contains only carbon and hydrogen gives 330 g of CO_2 and 270 g of water on complete combustion. The percentage of carbon and hydrogen, respectively are:
 (A) 25 and 75 (B) 40 and 60 (C) 60 and 40 (D) 75 and 25
22. Which one of the following is a water soluble vitamin, that is not excreted easily?
 (A) Vitamin B_2 (B) Vitamin B_1 (C) Vitamin B_6 (D) Vitamin B_{12}
23. A commercially sold conc. HCl is 35% HCl by mass. If the density of this commercial acid is 1.46 g/mL, the molarity of this solution is:
 (Atomic Mass: $\text{Cl} = 35.5$ amu, $\text{H} = 1$ amu)
 (A) 10.2 M (B) 12.5 M (C) 14.0 M (D) 18.2 M
24. Compound A contains 8.7% Hydrogen, 74% Carbon and 17.3% Nitrogen. The molecular formula of the compound is _____.
 [Given: Atomic masses of C, H and N are 12, 1 and 14 amu respectively. The molar mass of the compound A is 162 g mol^{-1}]
 (A) $\text{C}_4\text{H}_6\text{N}_2$ (B) $\text{C}_2\text{H}_3\text{N}$ (C) $\text{C}_5\text{H}_7\text{N}$ (D) $\text{C}_{10}\text{H}_{14}\text{N}_2$
25. Using the rules for significant figures, the correct answer for the expression $\frac{0.02858 \times 0.112}{0.5702}$ will be:
 (A) 0.005613 (B) 0.00561 (C) 0.0056 (D) 0.006
26. Production of iron in blast furnace follows the following equation
 $\text{Fe}_3\text{O}_4(\text{s}) + 4\text{CO}(\text{g}) \rightarrow 3\text{Fe}(\text{l}) + 4\text{CO}_2(\text{g})$
 when 4.640 kg of Fe_3O_4 and 2.520 kg of CO are allowed to react then the amount of iron (in g) produced is:
 [Given: Molar Atomic mass (g mol^{-1}): $\text{Fe} = 56$
 Molar Atomic mass (g mol^{-1}): $\text{O} = 16$
 Molar Atomic mass (g mol^{-1}): $\text{C} = 12$]
 (A) 1400 (B) 2200 (C) 3360 (D) 4200
27. The compound(s) that is(are) removed as slag during the extraction of copper is:
 (1) CaO (2) FeO (3) Al_2O_3 (4) ZnO
 (5) NiO
 Choose the correct answer from the options given below:
 (A) (3) (4) Only (B) (1), (2), (5) Only (C) (1), (2) Only (D) (2) Only
28. Haemoglobin contains 0.34% of iron by mass. The number of Fe atoms in 3.3 g of haemoglobin is:
 (Given: Atomic mass of Fe is 56 u, N_A in $6.022 \times 10^{23} \text{ mol}^{-1}$)
 (A) 1.21×10^5 (B) 12.0×10^{16} (C) 1.21×10^{20} (D) 3.4×10^{22}
29. Consider the above reaction, the limiting reagent of the reaction and number of moles of NH_3 formed respectively are:
 (A) H_2 , 1.42 moles (B) H_2 , 0.71 moles (C) N_2 , 1.42 moles (D) N_2 , 0.71 moles

30. 116 g of a substance upon dissociation reaction, yields 7.5 g of hydrogen, 60g of oxygen and 48.5g of carbon. Given that the atomic masses of H, O and C are 1, 16 and 12 respectively. The data agrees with how many formulae of the following?
 (A) CH_3COOH (B) HCHO (C) CH_3OOCH_3 (D) CH_3CHO
31. If a rocket runs on a fuel ($\text{C}_{15}\text{H}_{30}$) and liquid oxygen, the weight of oxygen required and CO_2 released for every litre of fuel respectively are: (Given: density of the fuel is 0.756 g/mL)
 (A) 1188 g and 1296 g (B) 2376 g and 2592 g
 (C) 2592g and 2376 g (D) 3429 g and 3142 g
32. Consider the reaction
 $4\text{HNO}_3(\text{l}) + 3\text{KCl}(\text{s}) \rightarrow \text{Cl}_2(\text{g}) + \text{NOCl}(\text{g}) + 2\text{H}_2\text{O}(\text{g}) + 3\text{KNO}_3(\text{s})$
 The amount of HNO_3 required to produce 110.0g of KNO_3 is:
 (Given: Atomic masses of H, O, N and K are 1, 16, 14 and 39, respectively.)
 (A) 32.2 g (B) 69.4 g (C) 91.5 g (D) 162.5 g
33. Complete combustion of 1.80 g of an oxygen containing compound ($\text{C}_x\text{H}_y\text{O}_z$) gave 2.64 g of CO_2 and 1.08 g of H_2O . The percentage of oxygen in the organic compound is:
 (A) 51.63 (B) 63.53 (C) 53.33 (D) 50.33

INTEGER TYPE QUESTIONS

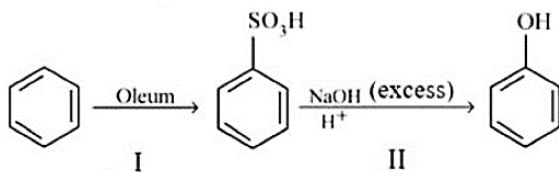
34. A protein 'A' contains 0.30% of glycine (molecular weight 75). The minimum molar mass of the protein 'A' is _____ $\times 10^3 \text{ g mol}^{-1}$ [nearest integer]
35. The number of N atoms is 681 g of $\text{C}_7\text{H}_5\text{N}_3\text{O}_6$ is $x \times 10^{21}$. The value of x is _____.
 ($N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$) (Nearest Integer)
36. 1 L aqueous solution of H_2SO_4 contains 0.02 m mol H_2SO_4 . 50% of this solution is diluted with deionized water to give 1 L solution (A). In solution (A), 0.01 m mol of H_2SO_4 are added. Total m mols of H_2SO_4 in the final solution is _____ $\times 10^3 \text{ m mols}$.
37. CNG is an important transportation fuel. When 100 g CNG is mixed with 208 oxygen in vehicles, it leads to the formation of CO_2 and H_2O and produces large quantity of heat during this combustion, then the amount of carbon dioxide, produced in grams is _____. [Nearest Integer]
 [Assume CNG to be methane]
38. The moles of methane required to produce 81 g of water after complete combustion is _____ $\times 10^{-2} \text{ mol}$. [Nearest Integer]
39. Two elements A and B which form 0.15 moles of A_2B and AB_3 type compounds. If both A_2B and AB_3 weigh equally, then the atomic weight of A is _____ times of atomic weight of B.
40. 56.0 L of nitrogen gas is mixed with excess of hydrogen gas and it is found that 20 L of ammonia gas is produced. The volume of unused nitrogen gas is found to be _____ L.

41. Chlorophyll extracted from the crushed green leaves was dissolved in water to make 2 L solution of Mg of concentration 48 ppm. The number of atoms of Mg in this solution is $x \times 10^{20}$ atoms. The value of x is _____. (Nearest Integer)
(Given: Atomic mass of Mg is 24 g mol^{-1} , $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$)



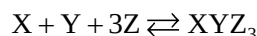
In the above reaction, 5 g of toluene is converted into benzaldehyde with 92% yield. The amount of benzaldehyde produced is $\text{_____} \times 10^{-2} \text{ g}$.
(Nearest integer)

43. In the following reaction,



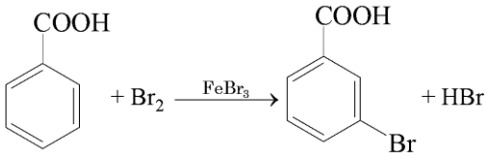
The % yield for reaction I is 60% and that of reaction II is 50%. The overall yield of the complete reaction is _____% [nearest integer]

44. In the given reaction,



if one mole of each of X and Y with 0.05 mol of Z gives compound XYZ_3 . (Given: Atomic masses of X, Y and Z are 10, 20 and 30 amu, respectively). The yield of XYZ_3 is _____ g. (Nearest integer)

45. On complete combustion of 0.492 g of an organic compound containing C, H and O, 0.7938 g of CO_2 and 0.4428 g of H_2O was produced. The % composition of oxygen in the compound is _____.
46. 2L of 0.2 M H_2SO_4 is reacted with 2L of 0.1 M NaOH solution, the molarity of the resulting product Na_2SO_4 in the solution is _____ millimolar. (Nearest integer).
47. A 1.84 mg sample of polyhydric alcoholic compound 'X' of molar mass 92.0 g/mol gave 1.344 mL of H_2 gas at STP. The number of alcoholic hydrogens present in compound 'X' is _____.
48. The formula of a gaseous hydrocarbon which requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y . The value of y is _____.
49. When 35 mL of 0.15 M lead nitrate solution is mixed with 20 mL of 0.12 M chromic sulphate solution, _____ $\times 10^{-5}$ moles of lead sulphate precipitate out. (Round off to the Nearest Integer).
50. The number of significant figures in 50000.020×10^{-3} is _____.

51. The number of chlorine atoms in 20 mL of chlorine gas at STP is ____ 10^{21} . (Round off to the Nearest Integer).
[Assume chlorine is an ideal gas at STP
 $R = 0.083 \text{ L bar mol}^{-1} \text{ K}^{-1}$, $N_A = 6.023 \times 10^{23}$]
52. 4g equimolar mixture of NaOH and Na_2CO_3 contains x g of NaOH and y g of Na_2CO_3 . The value of x is ____ g. (Nearest integer)
53. 250 mL of 0.5 M NaOH was added to 500 mL of 1 M HCl. The number of unreacted HCl molecules in the solution after complete reaction is ____ $\times 10^{21}$. (Nearest integer)
($N_A = 6.022 \times 10^{23}$)
54. The mole fraction of a solute in a 100 molal aqueous solution ____ $\times 10^{-2}$.
(Round off to the Nearest Integer).
[Given: Atomic masses: H: 1.0 u, O: 16.0 u]
55. 

Consider the above reaction where 6.1 g of benzoic acid is used to get 7.8 g of m-bromo benzoic acid. The percentage yield of the product is ____.
(Round off to the Nearest integer)
[Given: Atomic Masses: C = 12.0u, H : 1.0u, O : 16.0u, Br = 80.0 u]
56. A 6.50 molal solution of KOH (aq.) has a density of 1.89 g cm^{-3} . The molarity of the solution is ____ mol dm^{-3} .
(Round off to the Nearest Integer).
[Atomic masses: K :39.0 u; O :16.0 u; H :1.0 u]
57. The number of significant figures in 0.00340 is:
58. Complete combustion of 3 g of ethane gives $x \times 10^{22}$ molecules of water. The value of x is ____.
(Round off to the Nearest Integer).
[Use: $N_A = 6.023 \times 10^{23}$; Atomic masses in u : C : 12.0 ; O : 16.0 ; H : 1.0]
59. An aqueous KCl solution of density 1.20 g mL^{-1} has a molality of 3.30 mol kg^{-1} . The molarity of the solution in mol L^{-1} is ____.
(Nearest integer) [Molar mass of KCl = 74.5]
60. Consider the complete combustion of butane, the amount of butane utilized to produce 72.0 g of water is ____ $\times 10^{-1}$ g. (in nearest integer)
61. Sodium oxide reacts with water to produce sodium hydroxide. 20.0 g of sodium oxide is dissolved in 500 mL of water. Neglecting the change in volume, the concentration of the resulting NaOH solution is ____ $\times 10^{-1}$ M. (Nearest integer)
[Atomic mass: Na = 23.0, O = 16.0, H = 1.0]

62. If 80 g of copper sulphate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is dissolved in deionised water to make 5 L of solution. The concentration of the copper sulphate solution is $x \times 10^{-3} \text{ mol L}^{-1}$. The value of x is _____.
[Atomic masses Cu : 63.54 u, S : 32 u, O : 16 u, H : 1 u]
63. 100g of propane is completely reacted with 1000g of oxygen. The mole fraction of carbon dioxide in the resulting mixture is $x \times 10^{-2}$. The value of x is _____. (Nearest integer)
[Atomic weight: H = 1.008; C = 12.00; O = 16.00]

[JEE Main-2023]

64. The density of 3 M solution of NaCl is 0.1 g mL^{-1} . Molality of the solution is: _____ $\times 10^{-2} \text{ m}$ (Nearest Integer)
Given; Molar mass of Na and Cl is 23 and 35.5 g mol^{-1} respectively.
65. Some amount of dichloromethane (CH_2Cl_2) is added to 671.141 mL of chloroform (CHCl_3) to prepare $2.6 \times 10^{-3} \text{ M}$ solution of CH_2Cl_2 (DCM). The concentration of DCM is _____ ppm (by mass).
Given: atomic mass: C = 12; H = 1; Cl = 35.5
density of $\text{CHCl}_3 = 1.49 \text{ g cm}^{-3}$
66. A 300 mL bottle of soft drink has 0.2 M CO_2 dissolved in it. Assuming CO_2 behaves as an ideal gas, the volume of the dissolved CO_2 at STP is _____ mL. (Nearest integer)
Given: At STP, molar volume of an ideal gas is 22.7 L mol^{-1}
67. The molality of a 10%(v/v) solution of di-bromine solution in CCl_4 (carbon tetrachloride) is 'x'.
 $x = \times 10^{-2} \text{ M}$. (Nearest integer)
[Given: molar mass of $\text{Br}_2 = 160 \text{ g mol}^{-1}$
atomic mass of C = 12 g mol^{-1}
atomic mass of Cl = 35.5 g mol^{-1}
density of dibromine = 3.2 g cm^{-3}
density of $\text{CCl}_4 = 1.6 \text{ g cm}^{-3}$]
68. The number of units, which are used to express concentration of solutions from the following is _____
Mass percent. Mole, Mole fraction. Molarity, ppm. Molality
69. Number of hydrogen atoms per molecule of a hydrocarbon A having 85.8 % carbon is _____.
(Given: Molar mass of A = 84 g mol^{-1})
70. The volume of HCl, containing 73 g L^{-1} , required to completely neutralise NaOH obtained by reacting 0.69 g of metallic sodium with water, is _____ mL. (Nearest Integer)
(Given: molar Masses of Na, Cl, O, H, are 23, 35.5, 16 and 1 g mol^{-1} respectively)

71. When 0.01 mol of an organic compound containing 60% carbon was burnt completely, 4.4 g of CO_2 was produced. The molar mass of compound is _____ g mol^{-1} (Nearest integer).
72. When a hydrocarbon A undergoes complete combustion it requires 11 equivalents of oxygen and produces 4 equivalents of water. What is the molecular formula of A ?
 (A) C_{11}H_4 (B) C_{11}H_8
 (C) C_9H_8 (D) C_5H_8

[JEE Main-2024]

73. Consider the following reaction:
 $3\text{PbCl}_2 + 2(\text{NH}_4)_3\text{PO}_4 \rightarrow \text{Pb}_3(\text{PO}_4)_2 + 6\text{NH}_4\text{Cl}$
 If 72 mmol of PbCl_2 is mixed with 50 mmol of $(\text{NH}_4)_3\text{PO}_4$, then the amount of $\text{Pb}_3(\text{PO}_4)_2$ formed is _____ mmol (nearest integer).
74. 10 mL of gaseous hydrocarbon on combustion gives 40 mL of CO_2 (g) and 50 mL of water vapour. Total number of carbon and hydrogen atoms in the hydrocarbon is _____.
75. Following Kjeldahl's method, 1 g of organic compound released ammonia, that neutralised 10 mL of $2\text{M H}_2\text{SO}_4$. The percentage of nitrogen in the compound is _____%.
76. The quantity which changes with temperature is:
 (1) Molality (2) Mole fraction (3) Molarity (4) Mass percentage
77. 9.3 g of aniline is subjected to reaction with excess of acetic anhydride to prepare acetanilide. The mass of acetanilide produced if the reaction is 100% completed is _____ $\times 10^{-1}\text{g}$.
 (Given molar mass in g mol^{-1} N : 14, O : 16,
 C : 12, H : 1)
78. The mass of sodium acetate (CH_3COONa) required to prepare 250 mL of 0.35M aqueous solution is _____ g.
 (Molar mass of CH_3COONa is 82.02 g mol^{-1})
79. A sample of CaCO_3 and MgCO_3 weighed 2.21 g is ignited to constant weight of 1.152 g. The composition of mixture is:
 (Given molar mass in g mol^{-1} CaCO_3 : 100, MgCO_3 : 84)
 (1) 1.187 g CaCO_3 + 1.187 g MgCO_3
 (2) 1.023 g CaCO_3 + 1.023 g MgCO_3
 (3) 1.023 g CaCO_3 + 1.187 g MgCO_3
 (4) 1.187 g CaCO_3 + 1.023 g MgCO_3

EXERCISE #4 (JEE ADVANCED)

1. Paragraph for Question Nos. (i) to (iii)

Chemical reactions involve interaction of atoms and molecules. A large number of atoms/molecules (approximately 6.023×10^{23}) are present in a few grams of any chemical compound varying with their atomic/molecular masses. To handle such large numbers conveniently, the mole concept was introduced. This concept has implications in diverse areas such as analytical chemistry, biochemistry, electrochemistry and radiochemistry. The following example illustrates a typical case, involving chemical / electrochemical reaction, which requires a clear understanding of the mole concept.

A 4.0 molar aqueous solution of NaCl is prepared and 500 mL of this solution is electrolysed. This leads to the evolution of chlorine gas at one of the electrodes (atomic mass: Na = 23, Hg = 200; 1 Faraday = 96500 coulombs).

[At the anode: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

At the cathode: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$

$\text{Na} + \text{Hg} \rightarrow \text{NaHg}$ (sodium amalgam)]

(These reactions were not present in IIT-JEE paper)

(i) The total number of moles of chlorine gas evolved is: [JEE-2007, 4/162]

- (A) 0.5 (B) 1.0 (C) 2.0 (D) 3.0

(ii) If the cathode is a Hg electrode, the maximum weight (g) of amalgam formed from this solution is:

[JEE-2007, 4/162]

- (A) 200 (B) 225 (C) 400 (D) 446

(iii) The total charge (coulombs) required for complete electrolysis is : [JEE-2007, 4/162]

- (A) 24125 (B) 48250 (C) 96500 (D) 193000

2. A student performs a titration with different burettes and finds titre values of 25.2 mL, 25.25 mL, and 25.0 mL. The number of significant figures in the average titre value is : [JEE 2010, 3/163]

3. Reaction of Br_2 with Na_2CO_3 in aqueous solution gives sodium bromide and sodium bromate with evolution of CO_2 gas. The number of sodium bromide molecules involved in the balanced chemical equation is: [JEE 2011, 4/180]

4. Dissolving 120 g of urea (mol. wt. 60) in 1000 g of water gave a solution of density 1.15 g/mL. The molarity of the solution is: [JEE 2011, 3/160]

- (A) 1.78 M (B) 2.00 M (C) 2.05 M (D) 2.22 M

5. 29.2% (w/w) HCl stock solution has a density of 1.25 g mL^{-1} . The molecular weight of HCl is 36.5 g mol^{-1} . The volume (mL) of stock solution required to prepare a 200 mL solution of 0.4 M HCl is:

[JEE 2012, 4/136]

6. For the reaction: $\text{I}^- + \text{ClO}_3^- + \text{H}_2\text{SO}_4 \longrightarrow \text{Cl}^- + \text{HSO}_4^- + \text{I}_2$

The correct statement(s) in the balanced equation is/are :

[JEE(Advanced) 2014, 3/120]

- (A) Stoichiometric coefficient of HSO_4^- is 6. (B) Iodide is oxidized.
(C) Sulphur is reduced. (D) H_2O is one of the products.

7. A compound H_2X with molar weight of 80 g is dissolved in a solvent having density of 0.4 g ml^{-1} . Assuming no change in volume upon dissolution, the **molality** of a 3.2 molar solution is:
[JEE(Advanced) 2014, 3/120]

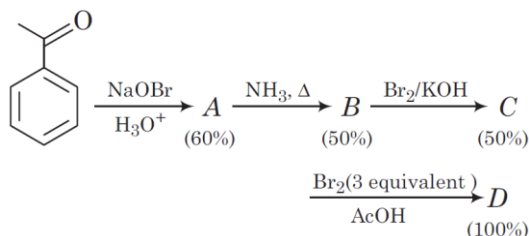
8. The mole fraction of a solute in a solution is 0.1. At 298 K, molarity of this solution is the same as its molality. Density of this solution at 298 K is 2.0 g cm^{-3} . The ratio of the molecular weights of the solute and solvent, $\left(\frac{m_{\text{solute}}}{m_{\text{solvent}}}\right)$ is
[JEE ADVANCE 2016]

9. Galena (an ore) is partially oxidised by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnace such that the content undergo self-reduction. The weight (in kg) of Pb produced per kg of O_2 consumed is
(Atomic weights in g mol^{-1} : O = 16, S = 32, Pb = 207)
[JEE ADVANCE 2018]

10. To measure the quantity of $MnCl_2$ dissolved in an aqueous solution, it was completely converted to $KMnO_4$ using the reaction, $MnCl_2 + K_2S_2O_8 + H_2O \rightarrow KMnO_4 + H_2SO_4 + HCl$ (equation not balanced).
Few drops of concentrated HCl were added to this solution and gently warmed. Further, oxalic acid (225 mg) was added in portions till the colour of the permanganate ion disappeared. The quantity of $MnCl_2$ (in mg) present in the initial solution is
(Atomic weights in g mol^{-1} : Mn = 55, Cl = 35.5)
[JEE ADVANCE 2018]

11. The ammonia prepared by treating ammonium sulphate with calcium hydroxide is completely used by $NiCl_2 \cdot 6H_2O$ to form a stable coordination compound. Assume that both the reactions are 100% complete. If 1584 g of ammonium sulphate and 952 g of $NiCl_2 \cdot 6H_2O$ are used in the preparation, the combined weight (in grams) of gypsum and the nickel-ammonia coordination compound thus produced is ____.
(Atomic weights in g mol^{-1} : H = 1, N = 14, O = 16, S = 32, Cl = 35.5, Ca = 40, Ni = 59)
[JEE ADVANCE 2018]

12. In the following reaction sequence, the amount of D (in gram) formed from 10 moles of acetophenone is.....
(Atomic weights in g mol^{-1} : H = 1, C = 12, N = 14, O = 16, Br = 80. The yield (%) corresponding to the product in each step is given in the parenthesis)
[JEE ADVANCE 2018]



13. Aluminium reacts with sulphuric acid to form aluminium sulphate and hydrogen. What is the volume of hydrogen gas in litre (L) produced at 300 K and 1.0 atm pressure, when 5.4 g of aluminium and 50.0 mL of 5.0 M sulphuric acid are combined for the reaction?
(Use molar mass of aluminium as 27.0 g mol^{-1} , $R = 0.082 \text{ atm L mol}^{-1} \text{ K}^{-1}$)
[JEE ADVANCE 2020]

Answer Key**Do yourself-1:**

1. N_A 2. (C)
3. (i) $H = 4N_A$, $S = 2N_A$, $O = 8N_A$ atoms (ii) $H = 4$ atoms, $S = 2$ atoms, $O = 8$ atoms.
 (iii) $H = 10N_A$, $S = 10N_A$, $O = 40 N_A$ atoms (iv) $H = 6$ atoms, $S = 6$ atoms, $O = 18$ atoms.
4. 1.88×10^{22} 5. (C) 6. (A) 7. (B)

Do yourself-2:

1. (A) 2. 2.16 g 3. (C) 4. (C)
5. (C)

Do yourself-3:

1. (i) O_2 is left in excess.
 (ii) 3 moles of O_2 or 96 g of O_2 is left.
 (iii) 2 moles of CO or 56 g of CO is formed.
 (iv) To use O_2 completely, total 8 moles of carbon or 96 g of carbon is needed.
2. (A) 3. (A) 4. (A) 5. (C)

Do yourself-4:

1. 35.33% 2. (D) 3. (A) 4. (A)
5. 11 6. 0.95 g

Do yourself-5:

1. (D) 2. (72) 3. (B) 4. (B)
5. (A) 6. (A) 7. (D) 8. (A)
9. (C)

Do yourself-6:

1. (D) 2. (B) 3. (B) 4. (D)
5. (20%) 6. (B)

Do yourself-7:

1. (A) 2. (A) 3. (B) 4. (B)
5. (A)

EXERCISE #1**Dalton's Atomic Theory**

1. (D) 2. (C) 3. (B)

Laws of Chemical Combination

1. (B) 2. (A) 3. (C) 4. (D)
5. (A) 6. (B) 7. (B) 8. (C)
9. (C) 10. (C) 11. (A) 12. (C)
13. (B) 14. (C) 15. (B)

Percentage Composition

- | | | | |
|--------|---------|--------|--------|
| 1. (B) | 2. (A) | 3. (A) | 4. (A) |
| 5. (A) | 6. (D) | 7. (C) | 8. (C) |
| 9. (C) | 10. (B) | | |

Mole Concept

- | | | | |
|---------|---------|---------|---------|
| 1. (C) | 2. (B) | 3. (C) | 4. (D) |
| 5. (C) | 6. (D) | 7. (B) | 8. (A) |
| 9. (B) | 10. (A) | 11. (C) | 12. (D) |
| 13. (B) | 14. (A) | 15. (C) | 16. (A) |
| 17. (D) | 18. (A) | 19. (B) | 20. (B) |
| 21. (C) | 22. (B) | 23. (A) | 24. (D) |
| 25. (B) | | | |

Limiting Reagent

- | | | | |
|--------|--------|--------|--------|
| 1. (B) | 2. (C) | 3. (C) | 4. (A) |
| 5. (B) | 6. (A) | 7. (B) | |

Concentration Terms

- | | | | |
|---------|---------|---------|---------|
| 1. (C) | 2. (A) | 3. (B) | 4. (B) |
| 5. (B) | 6. (D) | 7. (B) | 8. (B) |
| 9. (A) | 10. (A) | 11. (C) | 12. (A) |
| 13. (B) | 14. (B) | 15. (A) | 16. (B) |
| 17. (A) | 18. (B) | 19. (B) | 20. (D) |

EXERCISE #2

- | | | | |
|---------|---------|---------|---------|
| 1. (A) | 2. (B) | 3. (D) | 4. (C) |
| 5. (C) | 6. (A) | 7. (C) | 8. (D) |
| 9. (A) | 10. (C) | 11. (A) | 12. (D) |
| 13. (D) | 14. (C) | 15. (D) | 16. (A) |
| 17. (B) | 18. (C) | 19. (C) | 20. (D) |
| 21. (A) | 22. (D) | 23. (C) | 24. (A) |
| 25. (C) | 26. (B) | 27. (D) | 28. (C) |
| 29. (C) | 30. (D) | 31. (D) | 32. (C) |
| 33. (A) | 34. (C) | 35. (C) | 36. (D) |
| 37. (A) | 38. (B) | 39. (A) | 40. (D) |
| 41. (B) | 42. (C) | 43. (A) | 44. (D) |
| 45. (B) | 46. (C) | 47. (A) | 48. (D) |
| 49. (D) | 50. (D) | 51. (A) | 52. (A) |
| 53. (D) | 54. (D) | 55. (A) | 56. (A) |
| 57. (D) | 58. (B) | 59. (A) | 60. (B) |
| 61. (C) | 62. (B) | 63. (B) | 64. (C) |
| 65. (C) | 66. (D) | 67. (D) | 68. (D) |
| 69. (A) | 70. (B) | 71. (B) | 72. (B) |
| 73. (B) | 74. (A) | 75. (C) | 76. (B) |
| 77. (A) | 78. (B) | 79. (A) | 80. (B) |
| 81. (C) | 82. (C) | 83. (B) | 84. (C) |
| 85. (B) | | | |

EXERCISE #3 (JEE MAIN)

- | | | | |
|-----------|------------|------------|-----------|
| 1. (1) | 2. (3) | 3. (4) | 4. (1) |
| 5. (4) | 6. (2) | 7. (1) | 8. (2) |
| 9. (1) | 10. (2) | 11. (3) | 12. (4) |
| 13. (1) | 14. (3) | 15. (1) | 16. (4) |
| 17. (1) | 18. (3) | 19. (4) | 20. (4) |
| 21. (D) | 22. (D) | 23. (C) | 24. (D) |
| 25. (B) | 26. (C) | 27. (D) | 28. (C) |
| 29. (C) | 30. (B) | 31. (C) | 32. (C) |
| 33. (C) | 34. (25) | 35. (5418) | 36. (0) |
| 37. (143) | 38. (225) | 39. (2) | 40. (46) |
| 41. (24) | 42. (530) | 43. (30) | 44. (2) |
| 45. (46) | 46. (25) | 47. 3 or 6 | 48. (8) |
| 49. (525) | 50. (7) | 51. (1) | 52. (1) |
| 53. (226) | 54. (64) | 55. (78) | 56. (9) |
| 57. (4) | 58. (18) | 59. (3) | 60. (464) |
| 61. (13) | 62. (64) | 63. (19) | 64. (364) |
| 65. (221) | 66. (1362) | 67. (139) | 68. (5) |
| 69. (12) | 70. (15) | 71. (200) | 72. (C) |
| 73. (24) | 74. (14) | 75. (56) | 76. (3) |
| 77. (135) | 78. (7) | 79. (4) | |

EXERCISE #4 (JEE ADVANCED)

- | | | |
|---------------------------------|------------|------------|
| 1. (i)-(B), (ii)-(D), (iii)-(D) | 2. 3 | 3. 5 |
| 4. (C) | 5. 8 mL | 6. (A,B,D) |
| 8. 9 | 9. 6.47 kg | 7. 8 |
| 12. 495 g | 13. 6.15 | 10. 126 mg |
| | | 11. 2992 |



PW Web/App - <https://smart.link/7wwosivoicgd4>
 Library- <https://smart.link/sdfez8ejd80if>